

CPTG Geometric Pi Branch

Cosmological Comparison Coordinates

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Abstract

This article is the canonical technical reference for the Curvature Polarization Transport Gravity (CPTG) geometric pi branch and its comparison-coordinate system. CPTG contains one native cosmological branch, the geometric π branch. The acoustic and luminosity branches are projections of that native branch into their respective observational coordinates; they are not independent cosmologies or additional native expansion laws. The reference also records the interface between redshift-selected native phase and object-dependent bound-system structure, and introduces the symmetric acoustic–luminosity midpoint prescription $\omega_{\text{AL}} = 1/2$ for calculations that require a central bracket between those two projected coordinates. Its purpose is operational rather than evidentiary: it records the quantities, dependency chains, registry classes, transport maps, fixed inputs, derived values, implementation conventions, and sector-specific coordinate constructions used across CPTG cosmology, bound-system, and nuclear-reaction papers. A reader should be able to begin with the native branch, identify the correct upstream quantities for a target sector, apply the declared CPTG map in the required order, distinguish derived quantities from frozen inputs or data-conditioned coordinates, and locate the corresponding implementation and provenance record.

The guide is organized in the order in which the framework is used: native branch and scale architecture first; the cross-scale bound-system/native-phase interface second; quantity taxonomy and parameter registry third; then the nuclear/primordial, CMB/acoustic, perturbation, growth/lensing, distance, common-phase clock and horizon, BAO, and ShapeFit construction layers; followed by implementation checks, validation records, open conditions, and evidence identities. The validation sections document how the registered quantities have been exercised and where their current implementation boundaries lie; they are not used here as a proof of CPTG. Dedicated CPTG papers remain the sources for sector-specific derivations, tests, and scientific arguments, while this document supplies the common technical vocabulary and reference equations needed to keep those papers mutually consistent.

Contents

1	How to Use This Reference	4
2	Core Branch and Reference Conventions	5
3	Scale and Native-Phase Architecture	7
4	Bound-System Interface to Native Phase	8
5	Quantity Taxonomy and Parameter Registry	10
5.1	No-Retuning Rule for Comparison Coordinates	17
6	Microscopic Nuclear and Primordial BBN Comparison Layer	17
6.1	Commissioning-Channel Reaction-Observable Coordinates	18
6.2	Network-Conditioned Nuclear Reaction-Response Pushforward	20
6.3	Primordial Nuclear and Mass-Seven Abundance-Transport Coordinates	21
7	Locked CMB Acoustic Transport Layer	21
7.1	Geometric Origin of the CMB Amplitude and Visibility Gates	22
7.1.1	Amplitude Projection and the $\pi/3$ Curvature Gate	23
7.1.2	Visibility Transport and the $\mathcal{G}_T^2$ Reionization-Depth Gate	23
8	CMB Comparison Coordinates and Reference Quantities	23
9	Recombination-Scale Perturbation and CMB Projection Construction	24
9.1	Curvature-Transport Spectrum Bridge	24
9.2	Data-Conditioned CMB Map-Envelope Coordinate	25
10	Late-Time Growth and Lensing Construction	26
11	Large-Scale Distance, Horizon, and Acoustic-Ruler Comparison Layer	27
11.1	Acoustic-Branch Clock and Density-Weighted Diagnostic	28
11.2	Luminosity Projection, Clock, and Endpoint	29

11.3	Acoustic–Luminosity Midpoint Comparison Coordinate	30
11.4	Common-Phase Horizon and Last-Scattering Coordinates	32
12	Primordial Network Implementation and Verification Record	34
12.1	Commissioning-Channel Reaction-Observable Verification	34
12.2	PRIMAT-Native Baryon, Radiation, and Mass-Seven Validation	35
12.3	PRIMAT-Native First-Order Reaction Pushforward Validation	36
13	CMB Implementation and Full-Stack Verification Record	37
13.1	Curvature-Transport Spectrum Bridge Verification	38
13.2	CMB Map-Envelope Verification	38
13.3	Fixed-Row Likelihood Verification Protocol	38
13.4	Validation Milestone Summary	39
13.5	High- ℓ Profile Robustness	40
13.6	Plik and SPT Posterior-Proxy Stability	40
13.7	Low- ℓ and Lensing Diagnostic Stability	41
13.8	Combined-Stack Profile Validation	41
13.9	Combined-Stack Posterior Stability Diagnostic	42
13.10	Numerical Amplitude and Reionization-Depth Gate Closure	43
13.11	Coupled Gate Perturbation	44
13.12	CMB Implementation Boundary	44
14	DESI DR1 Adapter and Diagnostic Record	45
14.1	Compressed ShapeFit Coordinate Comparison	45
14.2	BAO Quarter-Ruler Coordinate Wrapper	46
14.3	Full-Shape Spectrum AP/Growth Diagnostic	47
15	Late-Time Growth, Lensing, Distance, and Clock Verification Record	49
15.1	Compressed Weak-Lensing and CMB-Lensing S_8 Comparison	49
15.2	Pantheon+ Supernova Distance-Shape Test	50

15.3 Age and Clock Consistency	51
16 Implementation Status and Scope Boundaries	52
16.1 Sector Status Matrix	52
17 Limitations and Open Closure Conditions	54
18 Likelihood, Software, and Evidence Provenance	54
19 Reference Summary and Usage Boundaries	56
20 Data, Code, and Evidence Availability	56
20.1 Bound-System Native-Phase and Resolved-Inference Evidence	57
20.2 PRIMAT Full-Coordinate Primordial-Network Evidence	57
20.3 PRIMAT First-Order Reaction-Response Evidence	58
20.4 PArthENoPE Second-Authority Transfer Evidence	58
20.5 CMB, Supernova, Lensing, and DESI Evidence	58
20.6 Evidence Identity and Distribution	59

1 How to Use This Reference

This document is intended to be used as a technical lookup and construction guide. It is the common reference layer for CPTG papers that need the geometric pi branch, its derived quantities, or one of its observational or network comparison coordinates. The recommended workflow is:

1. identify the physical sector or reported observable to be constructed;
2. begin from the native branch quantities and the scale/epoch architecture defined below;
3. for a bound system, keep the redshift-selected native quantities separate from the object’s locally inferred structural quantities before forming any cross-scale comparison;
4. use the parameter registry to determine the status of every required quantity before using it;
5. follow the sector-specific dependency chain in the order stated, without skipping directly from an observational value to a native CPTG quantity;
6. keep frozen comparison inputs and fixed implementation settings distinct from quantities derived from the CPTG branch;
7. apply the no-retuning rule unless the target paper explicitly defines a different diagnostic protocol; and

8. consult the later implementation, validation, and evidence sections only to determine how the construction has been exercised, what numerical authority was used, and what limitations remain open.

The equations and registry entries in the construction sections are the reference layer. Validation statistics, likelihood outputs, network residuals, and diagnostic scans document implementation behavior; they do not redefine the registered CPTG quantities. When a sector-specific CPTG paper gives a more detailed derivation or evidence analysis, that paper supplies the specialized argument while this guide supplies the common definitions and coordinate relationships.

2 Core Branch and Reference Conventions

CPTG comparison work begins from one fixed native cosmological branch and proceeds outward through declared projection maps. The single native branch is the geometric π branch. The acoustic and luminosity branches are projections of that native branch into different observational coordinates; neither is an independently evolving cosmology, and neither adds a second native expansion law. This dependency order is the primary operating convention of the reference guide. A comparison coordinate is the sector-specific language in which a native or transported CPTG quantity is expressed; it is not a replacement definition of the native theory.

The cosmological branch architecture is therefore

$$\mathfrak{B}_\pi \xrightarrow{\mathcal{P}_{\text{ac}}} \mathfrak{B}_{\text{ac}}, \quad \mathfrak{B}_\pi \xrightarrow{\mathcal{P}_{\text{lum}}} \mathfrak{B}_{\text{lum}}, \quad (1)$$

where $\mathfrak{B}_\pi$ is native and $\mathfrak{B}_{\text{ac}}$ and $\mathfrak{B}_{\text{lum}}$ are projected branches. All three use the native phase coordinate

$$N_\pi \equiv \ln a, \quad (2)$$

The subscript is retained throughout this reference to distinguish cosmological phase N_π from the post-solution Structural Mode N used in CPTG bound-system analyses. Each branch carries its own integrated time coordinate. For any branch b whose full rate history $H_b(a)$ is defined,

$$H_b dt_b = dN_\pi, \quad t_b(a) - t_b(a_{\text{ref}}) = \int_{a_{\text{ref}}}^a \frac{da'}{a' H_b(a')}. \quad (3)$$

Consequently, branch synchronization is internal: no common clock or cross-branch thermal synchronization is imposed. The accumulated separation from the native time coordinate is

$$\Delta t_{b\pi}(a) = \int_{a_{\text{ref}}}^a \frac{da'}{a'} \left[\frac{1}{H_b(a')} - \frac{1}{H_\pi(a')} \right], \quad (4)$$

so different projected rate histories generally accumulate clock separation as the native phase advances. The native thermal property

$$\Theta_{\pi T}(a) \equiv \frac{T(a)}{T_0} = 1 + z = a^{-1}, \quad T_0 = 2.72548 \text{ K}, \quad (5)$$

belongs to $\mathfrak{B}_\pi$; it is not an additional cosmological branch [1].

The comparison hierarchy is

$$\begin{array}{ccc}
\text{CPTG-native geometric quantity} & & \\
\downarrow & & \\
\text{comparison-coordinate quantity} & & (6) \\
\downarrow & & \\
\text{reported observational summary} & &
\end{array}$$

The first layer is fixed by the geometric branch. The second layer expresses the native result in the variables used by a particular observational pipeline. The third layer is the published or reconstructed quantity against which the comparison is made.

The native and projected branch values used here are

$$p_C = \pi, \quad (7)$$

$$p_{\text{ac}} \equiv 3 - \frac{\pi}{100} \simeq 2.968584073464. \quad (8)$$

The displayed twelve-decimal value is the locked numerical representation used by the comparison pipelines; the analytic definition remains $3 - \pi/100$.

$$\mathcal{G}_T \equiv \frac{p_{\text{ac}}}{p_C} \simeq 0.9449296585513721, \quad (9)$$

with rate factor

$$\sqrt{\mathcal{G}_T} \simeq 0.9720749243506758. \quad (10)$$

The native late and transport weights are

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}, \quad (11)$$

so the native background is $E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}$.

The acoustic projection is fixed at the background level by subtracting the native late term,

$$X_\pi(a) \equiv E_\pi(a)^2 - \mathcal{B}_\pi = \mathcal{T}_\pi a^{-\pi}, \quad (12)$$

and applying the exact normalized projector

$$\begin{aligned}
X_{\text{ac}}(a) &\equiv \mathcal{T}_\pi^{1-\mathcal{G}_T} X_\pi(a)^{\mathcal{G}_T} \\
&= \mathcal{T}_\pi a^{-p_{\text{ac}}},
\end{aligned} \quad (13)$$

$$E_{\text{ac}}(a)^2 = \mathcal{B}_\pi + X_{\text{ac}}(a). \quad (14)$$

The prefactor preserves the native normalization $X_{\text{ac}}(1) = X_\pi(1) = \mathcal{T}_\pi$; the exponent change is therefore a projection of the native transport term rather than a second native background.

The native geometric acceleration anchor is

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} = 5.568327996831708 \times 10^{-10} \text{ m s}^{-2}. \quad (15)$$

The native geometric normalization is

$$H_0^{(\pi)} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (16)$$

and the acoustic comparison normalization, also used by the CMB coordinate, is

$$H_{0,\text{ac}}^{\text{CPTG}} \equiv H_{0,\text{CMB}}^{\text{CPTG}} = H_0^{(\pi)} \sqrt{\mathcal{G}_T} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (17)$$

The framework and public computational materials are documented in Refs. [38, 47].

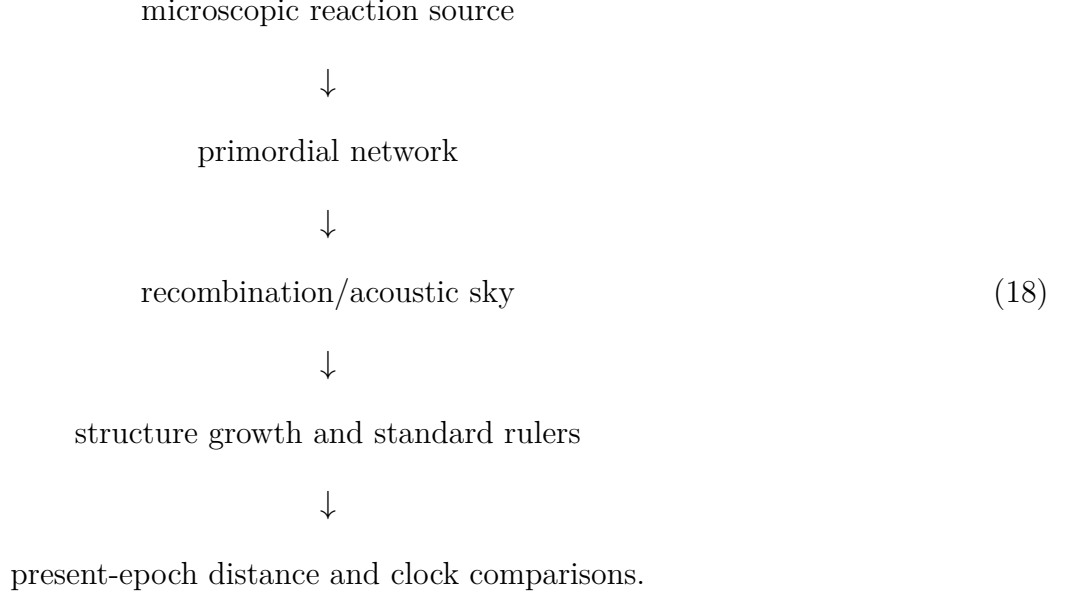
3 Scale and Native-Phase Architecture

CPTG is a framework of nested physical scales. The presentation therefore uses physical scale as its primary organizational axis, proceeding from microscopic nuclear structure to cosmological distance scales. Within each scale regime, the secondary ordering follows native geometric phase, equivalently the observed redshift coordinate, from the earliest validated phase toward the normalized present. Branch-specific times are obtained only after that phase has been selected and are not used as the ordering variable. The phrase “earliest phase” is used deliberately: the comparison program begins with the earliest phase for which the present paper supplies a defined and tested coordinate, not with a claim about the mathematical limit $t = 0$.

Regime	Physical scale	Epoch ordering	Comparison layers
I	Scale-independent geometric foundation and cross-scale interfaces	Applies across all validated native phases	Native pi branch, bound-system/native-phase interface, registry classes, and no-retuning comparison rule
II	Microscopic nuclear to primordial-network scale	Primordial BBN epoch	Reaction source and observable projection, native-network response, baryon/radiation transport, and live mass-seven abundance transport
III	Recombination and acoustic-horizon scale	Recombination and CMB observation	Acoustic transport, amplitude/visibility gates, perturbation-to-spectrum bridge, map-envelope coordinate, CMB likelihood closure, and common-phase horizon containment
IV	Structure-growth to cosmological-distance scale	Post-recombination evolution to the present epoch	Growth and lensing, distance/ruler coordinates, DESI BAO/full-shape diagnostics, supernova distance shape, and clock/age comparison

Table 1: Scale–phase organization used in this article. Physical scale is the primary reading axis; native geometric phase is the secondary axis within each regime.

The resulting conceptual progression is



Coordinates that bridge two regimes are placed at the transition where their physical interpretation is clearest. CMB lensing, for example, is sourced by the CMB but accumulates through intervening structure, so its compressed late-time amplitude is treated as a bridge from the acoustic regime into the structure-growth regime rather than as a separate fundamental scale.

4 Bound-System Interface to Native Phase

The cosmological branch identifies when an observation is represented on CPTG native geometry; it does not by itself determine the structural state of a bound object observed at that phase. This distinction is important whenever the comparison framework is applied to galaxies, clusters, or compact systems. Redshift selects the native phase N_π and therefore the native quantities $E_\pi(a)$, $a_\pi(a)$, and $R_C(a)$. The bound system supplies its own object-dependent structural acceleration $a_\star$ through the appropriate structural or dynamical reduction [38, 45].

For the positive structural branch,

$$R_c = \frac{48\pi c^2}{a_\star}, \quad a_\pi(a) = a_{\pi,0} E_\pi(a), \quad R_C(a) = \frac{48\pi c^2}{a_\pi(a)}. \tag{19}$$

Here lowercase R_c is the bound-system structural curvature scale and uppercase R_C is the native cosmological curvature scale. Neither is the visible radius of the gravitating object. At the continuous endpoint $a_\star = 0$, the structural scale satisfies $R_c \rightarrow \infty$ while $a_\star$ itself remains the regular variable.

Whenever $a_\pi > 0$, the cross-scale acceleration ratio has the continuous definition

$$\eta_\star \equiv \frac{a_\star}{a_\pi}. \tag{20}$$

For $a_\star > 0$, the common geometric numerator additionally gives

$$\eta_\star = \frac{R_C}{R_c}. \tag{21}$$

Thus $a_\star = 0$ implies $\eta_\star = 0$ by direct evaluation of the acceleration ratio, while the radius form is understood by the continuous limit $R_c \rightarrow \infty$. Native phase selection alone does not require $a_\star$ to follow a_π and does not rescale the local bound-system solution.

The structural acceleration can also be written as the associated CPTG structural-rate coordinate

$$H_{\text{struct}} \equiv \frac{16\pi}{3} \frac{a_\star}{c}, \quad (22)$$

while the native branch satisfies

$$H_\pi(a) = \frac{a_\pi(a)}{c\sqrt{\mathcal{B}_\pi}}. \quad (23)$$

Their dimensionless cross-scale rate ratio is therefore

$$\begin{aligned} \mathcal{K}_\star &\equiv \frac{H_{\text{struct}}}{H_\pi} \\ &= \frac{16\pi}{3} \sqrt{\mathcal{B}_\pi} \eta_\star \\ &= 13.8338192981 \eta_\star. \end{aligned} \quad (24)$$

Accordingly, $\mathcal{K}_\star$ is a derived cross-scale rate coordinate and carries no independent degree of freedom beyond $\eta_\star$. At $a_\star = 0$, $H_{\text{struct}} = 0$ and $\mathcal{K}_\star = 0$ continuously.

Define the native transport fraction

$$f_\pi(a) \equiv \frac{\mathcal{T}_\pi a^{-\pi}}{\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}}. \quad (25)$$

Because

$$\frac{d \ln E_\pi}{d N_\pi} = -\frac{\pi}{2} f_\pi(a), \quad (26)$$

the positive-branch logarithmic state–phase relation is

$$\frac{d \ln \eta_\star}{d N_\pi} = \frac{d \ln a_\star}{d N_\pi} + \frac{\pi}{2} f_\pi(a), \quad a_\star > 0. \quad (27)$$

The logarithmic form is not evaluated at $a_\star = 0$; the unnormalized quantities above provide the continuous endpoint.

Let $\mathcal{S}$ denote the intrinsic solved structural state of the bound system: the source distribution, geometry, support state, inferred $a_\star$, and corresponding solved local field. The recent NMBS-C7447 comparison isolates two distinct comparison operations based on Eq. (27) [43, 44, 45]. Under a strict fixed-local-state relocation,

$$\left. \frac{\partial a_\star}{\partial N_\pi} \right|_{\mathcal{S}} = 0, \quad (28)$$

so the source, $a_\star$, R_c , solved local field, and any post-solution Structural Mode remain fixed while the native quantities move with phase. Under a native-locked comparison trajectory,

$$\frac{d \ln \eta_\star}{d N_\pi} = 0, \quad (29)$$

which instead requires

$$\frac{d \ln a_\star}{dN_\pi} = -\frac{\pi}{2} f_\pi(a) = \frac{d \ln E_\pi}{dN_\pi}. \quad (30)$$

Because $\mathcal{K}_\star$ is proportional to $\eta_\star$, native locking also gives $d \ln \mathcal{K}_\star / dN_\pi = 0$, while fixed local structure lets both cross-scale ratios follow the native phase response. The two operations are therefore mathematically distinct. A native-locked comparison changes $a_\star$ and R_c and consequently requires the local CPTG field to be re-solved; a fixed-local-state comparison does not. Neither continuation is implied merely by assigning the observation a native phase.

Actual physical evolution is a third operation. In general one may write

$$a_\star = \mathcal{A}[\mathcal{S}], \quad \mathcal{S} = \mathcal{S}(N_\pi), \quad (31)$$

so the structural acceleration changes only insofar as the evolving bound-system state changes. Native locking could describe a special physical trajectory if the solved dynamics enforce it, but it is not imposed by the native branch itself.

The same distinction applies to measurement interfaces. The quantity $a_\star$ is the object-dependent structural acceleration regardless of how it is inferred. A one-radius pressure-support expression such as $\sigma_\star^2/R_e$ is a compressed estimator when only integrated kinematics are available; it is not the universal definition of $a_\star$. When resolved multi-radius dynamics are available, a field-selected multi-radius reconstruction supersedes the one-radius estimate as the more informative measurement of the same structural quantity. The inferred cross-scale quantities $\eta_\star$ and $\mathcal{K}_\star$ therefore inherit the measurement interface used for $a_\star$: one-radius and resolved values should not be pooled as though they have equivalent information content, and resolved solutions must retain their declared support, anisotropy, and geometry assumptions [46]. Re-expression of a published galaxy radius or mass through acoustic or luminosity distance coordinates is likewise a coordinate-sensitivity calculation; it does not by itself evolve or redefine the object’s intrinsic $a_\star$.

Quantity	Defining relation	Class	Role / boundary
$a_\star$	Object-dependent structural acceleration from the declared bound-system solve or estimator	Bound-system structural quantity	Same physical quantity across measurement interfaces; not set by redshift alone
R_c	$48\pi c^2/a_\star$ for $a_\star > 0$	Derived structural quantity	Bound-system curvature scale; $R_c \rightarrow \infty$ as $a_\star \rightarrow 0$
$\eta_\star$	$a_\star/a_\pi = R_C/R_c$ on the positive structural branch	Derived cross-scale coordinate	Acceleration/radius comparison; not phase invariant unless a continuation rule imposes it
H_{struct}	$(16\pi/3)(a_\star/c)$	Derived structural quantity	Rate representation of the same structural scale
$\mathcal{K}_\star$	$H_{\text{struct}}/H_\pi = (16\pi/3)\sqrt{\mathcal{B}_\pi} \eta_\star$	Derived cross-scale coordinate	Rate-space representation of $\eta_\star$; no independent degree of freedom

Table 2: Bound-system quantities and their interface to the native phase. The registry distinguishes the structural quantity itself from its derived cross-scale coordinates and from the observational method used to infer it.

5 Quantity Taxonomy and Parameter Registry

The comparison layer uses several different kinds of quantities. They are deliberately classified here so that a geometric identity, a derived comparison coordinate, a frozen pipeline input, and a numerical diagnostic are not all described as “locked parameters.” The classes used throughout the article are:

- **native invariant:** fixed by the CPTG geometric branch;

- **derived branch quantity**: calculated algebraically or by a stated branch integral from native invariants;
- **bound-system structural quantity**: an object-dependent structural quantity selected or inferred by the declared bound-system reduction, field solve, or boundary-value problem rather than by cosmological phase alone;
- **derived structural quantity**: an algebraic consequence of a declared bound-system structural quantity, retaining that object’s structural provenance;
- **derived comparison coordinate**: an arithmetic or integral consequence of a declared comparison map, expressed in an observational coordinate;
- **registered comparison prescription**: a fixed sector map stated and frozen before evaluation; its downstream arithmetic consequences may be derived comparison coordinates, but an independent upstream field/operator derivation is not implied unless explicitly cited;
- **transported comparison coordinate**: a fixed CPTG transport map applied to a frozen external comparison normalization;
- **bridge-validated comparison coordinate**: a comparison-level amplitude or power mapping whose coordinate response has been locked through the declared bridge validation, without claiming a completed native perturbation/source implementation;
- **data-conditioned comparison coordinate**: an observational map or operator whose definition explicitly includes a declared observed-data or external comparison scaffold; that scaffold is held fixed and is not promoted into CPTG-native theory;
- **abundance-transport comparison coordinate**: a fixed CPTG transport quantity applied dynamically inside a native abundance network, distinct from both the four-sector reaction simplex and any fitted endpoint correction;
- **network-conditioned comparison coordinate**: a comparison vector whose geometric source is fixed by CPTG but whose numerical pushforward is supplied by the declared native reaction-network dynamics and observable set; no code-independent static endpoint matrix is assumed;
- **composite comparison tuple**: an ordered collection of comparison quantities used together by a pipeline or native network; each entry retains its own registry class and the tuple does not erase those distinctions;
- **frozen comparison input**: a numerical input fixed before the corresponding likelihood or pipeline evaluation, but not promoted to a geometric invariant;
- **fixed comparison setting**: a value imposed by the declared comparison or adapter protocol and held fixed during evaluation, but neither derived as a CPTG coordinate nor fitted as a new theory parameter;
- **pipeline result**: a numerical output of a specified observational or likelihood calculation;
- **diagnostic / open derivation**: a useful comparison object whose independent geometric derivation is not yet promoted.

Here *native invariant* means theory-fixed by the single geometric π -branch definition before observational conversion. A projected-branch parameter may be an exact analytic consequence of π without becoming a

second native invariant. The registry therefore distinguishes the native curvature exponent $p_C = \pi$ from the derived acoustic-projection exponent $p_{ac} = 3 - \pi/100$.

The registry class also fixes the strongest permissible claim verb. This prevents a transported, data-conditioned, or diagnostic quantity from acquiring native-theory language merely because it appears in the same comparison table.

Registry class	Preferred claim language	Boundary
Native invariant	fixed	Geometric branch identity
Derived branch quantity	derived	Algebraic or integral consequence of native invariants
Bound-system structural quantity	inferred / selected by the declared structural solve	Object-specific structural quantity; measurement interface and support assumptions remain explicit
Derived structural quantity	derived from the named structural quantity	Algebraic structural consequence; not promoted to a native cosmological invariant
Derived comparison coordinate	mapped / predicted in the declared coordinate	Arithmetic consequence of the declared comparison map; not a new native invariant
Registered comparison prescription	registered / applied	Fixed mapping rule; independent native/operator derivation is not implied unless cited
Transported comparison coordinate	transported from the named frozen anchor	Anchor provenance remains explicit
Bridge-validated comparison coordinate	bridge-validated	Native source/Boltzmann closure is not implied
Data-conditioned comparison coordinate	constructed / evaluated conditional on the declared scaffold	Observed or external scaffold is not promoted into CPTG theory
Abundance-transport comparison coordinate	inserted / evolved in the native abundance network	Not an endpoint-fit factor or reaction-simplex coordinate
Network-conditioned comparison coordinate	source-predicted and network-pushed-forward	Native network dynamics remain authoritative for the pushforward
Composite comparison tuple	assembled from declared entries	Each entry retains its own registry class and provenance
Frozen comparison input	held fixed	Upstream numerical input, not a native invariant
Fixed comparison setting	imposed / fixed by protocol	Neither derived nor fit as a theory coordinate
Pipeline result	returned / measured by the named pipeline	Numerical evidence, not a theory parameter
Diagnostic / open derivation	diagnostically supported	Independent derivation or full validation remains open

Table 3: Claim-language rule associated with the paper-wide quantity taxonomy.

A survey-specific comparison can be written schematically as

$$\Theta_\pi \longrightarrow m_s(\Theta_\pi, \nu_s), \quad (32)$$

where Θ_π denotes the fixed CPTG branch, m_s is the model vector in the observational coordinate of sector s , and ν_s contains declared survey, pipeline, nuisance, observed-data, or external comparison-scaffold quantities. The comparison protocol permits these observational quantities to define the coordinate system without relabeling them as CPTG-native parameters [42].

Quantity	Defining relation or value	Class	Role
p_C	π	Native invariant	Curvature exponent
p_{ac}	$3 - \pi/100 \simeq 2.968584073464$	Derived branch quantity	Acoustic-projection exponent
$\mathcal{B}_\pi$	$1 - 1/\pi \simeq 0.681690113816$	Native invariant	Late/geometric weight
$\mathcal{T}_\pi$	$1/\pi \simeq 0.318309886184$	Native invariant	Transport weight
$\mathcal{G}_T$	$p_{ac}/p_C \simeq 0.9449296585513721$	Derived branch quantity	Transport factor
$\sqrt{\mathcal{G}_T}$	0.9720749243506758	Derived branch quantity	Hubble-rate transport
$\mathcal{G}_T^2$	0.8928920596100127	Derived branch quantity	Two-step visibility gate
$a_{\pi,0}$	$\pi^{3/2}10^{-10} = 5.568327996831708 \times 10^{-10} \text{ m s}^{-2}$	Native invariant	Acceleration anchor
N_π	$\ln a$	Derived branch quantity	Native cosmological phase; plain N is reserved for bound-system Structural Mode
$a_\pi(a)$	$a_{\pi,0}E_\pi(a)$	Derived branch quantity	Native acceleration at the selected phase
H_∞	$a_{\pi,0}/c = 57.3131992073 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Derived branch quantity	Asymptotic Hubble scale
$H_0^{(\pi)}$	$H_\infty/\sqrt{\mathcal{B}_\pi} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Derived branch quantity	Raw present normalization
R_{C0}	$48\pi c^2/a_{\pi,0} = 788.7823091477 \text{ Gpc}$	Derived branch quantity	Present native curvature scale
$R_C(a)$	$R_{C0}/E_\pi(a)$	Derived branch quantity	Native curvature-radius evolution
$t_{0,\pi}$	$12.754121032276 \text{ Gyr}$	Derived branch quantity	Present native branch time
$\Theta_{\pi T}(a)$	$T/T_0 = 1 + z = a^{-1}, T_0 = 2.72548 \text{ K}$	Derived branch quantity	Native thermal property; not a separate branch

Table 4: Native and directly derived geometric quantities. Decimal values are displayed numerical representations of the analytic definitions.

Quantity	Defining relation or value	Class	Role
$E_\pi(a)^2$	$\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}$	Derived branch quantity	Native expansion
$E_{\text{ac}}(a)^2$	$\mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}$	Derived branch quantity	Exact acoustic projection
$H_{0,\text{ac}}^{\text{CPTG}} \equiv H_{0,\text{CMB}}^{\text{CPTG}}$	$H_0^{(\pi)} \sqrt{\mathcal{G}_T} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Derived comparison coordinate	Acoustic/CMB Hubble normalization
$H_{\infty,\text{ac}}^{\text{CPTG}} \equiv H_{\infty,\text{CMB}}^{\text{CPTG}}$	$H_\infty \sqrt{\mathcal{G}_T} = 55.7127237837 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Derived comparison coordinate	Acoustic/CMB asymptote
$\Omega_{m,\text{ac}}$	$p_{\text{ac}}/(3\pi) \simeq 0.314976552850$	Derived comparison coordinate	Acoustic comparison weight
$\Omega_{\Lambda,\text{ac}}$	$1 - p_{\text{ac}}/(3\pi) \simeq 0.685023447150$	Derived comparison coordinate	Complementary acoustic comparison weight
$t_{0,\text{ac}}^{\text{CPTG}}$	13.88517358 Gyr, from the acoustic-projection age integral	Derived comparison coordinate	Acoustic branch time coordinate
$E_{\text{ac},\rho}(a)^2$	$\Omega_{\Lambda,\text{ac}} + \Omega_{m,\text{ac}} a^{-p_{\text{ac}}}$	Diagnostic / open derivation	Density-weighted acoustic overlay; not a branch
$t_{0,\text{ac},\rho}^{\text{CPTG}}$	13.926455176722689 Gyr, from the acoustic-density diagnostic integral	Diagnostic / open derivation	Density-weighted acoustic age diagnostic; not a branch clock
$\mathcal{L}_{\text{loc}}$	$1.054075942846 \pm 0.012389029805$	Data-conditioned comparison coordinate	Local luminosity-transfer coordinate; four-anchor SH0ES conditioned
$\Xi_{\text{loc}}(1)$	$0.948698247775 \pm 0.011150478244$	Data-conditioned comparison coordinate	Reciprocal present-epoch rate factor; endpoint of the conditioned luminosity clock map
$H_{0,\text{lum}}^{\text{CPTG}}$	$73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Data-conditioned comparison coordinate	SH0ES endpoint expressed in the luminosity projection
ω_{AL}	1/2	Registered comparison prescription	Symmetric acoustic-luminosity midpoint weight; not a third branch parameter
$\mathcal{M}_{\text{AL},0}$	$(1 - \omega_{\text{AL}})\sqrt{\mathcal{G}_T} + \omega_{\text{AL}}\mathcal{L}_{\text{loc}} = 1.013075433598 \pm 0.006194514903$	Data-conditioned comparison coordinate	Present rate-space realization of the midpoint prescription
$H_{0,\text{mid}}^{\text{CPTG}}$	$70.3238983675 \pm 0.43 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Data-conditioned comparison coordinate	Present acoustic-luminosity midpoint rate
$\Xi_{\text{lum}}^{\text{anchor}}(a)$	$H_\pi(a)/[H_{0,\text{lum}}^{\text{CPTG}} E_{\text{ac},\rho}(a)]$	Data-conditioned comparison coordinate	Phase-dependent SH0ES-conditioned luminosity clock map
$t_{0,\text{lum}}^{\text{anchor}}$	12.8430574232 Gyr	Data-conditioned comparison coordinate	Present luminosity time in the conditioned comparison realization
$\chi_\star$	13876.8219 Mpc at $z_\star = 1089.92$	Derived comparison coordinate	Acoustic comparison distance; native $H_0^{(\pi)}$ supplies the dimensional prefactor
$L_\star/R_C$	0.53792156 at $z_\star = 1089.92$	Derived comparison coordinate	Native radius-to-radius horizon-containment ratio
z_{enter}	3231.635355	Derived comparison coordinate	Common-phase entry where $L_\star = R_C$
$\mathcal{G}_L^{\text{candidate}}$	10/9	Diagnostic / open derivation	Candidate endpoint normalization; not required to define the luminosity comparison clock

Table 5: Background, clock, and distance-layer quantities.

Table 6: Baryon, radiation, primordial-network, nuclear-response, and CMB comparison-map quantities. Frozen inputs are fixed before evaluation but are not relabeled as native geometric invariants. The live mass-seven quantities are abundance-transport coordinates. The first-order nuclear response $\mathcal{C}_{r,\mathcal{N}}^{(1)}$ is network-conditioned because its pushforward is supplied by the declared native reaction dynamics, while the map-envelope operator $\mathcal{E}_\ell^M$ is data-conditioned because the observed harmonic scaffold is declared as part of the comparison coordinate.

Quantity	Relation or numerical value	Class	Role
f_b	$\pi/20 \simeq 0.157079632679$	Derived comparison coordinate	Baryon fraction
$\Omega_b h_{\text{ac}}^2$	0.0225278574944	Frozen comparison input	Acoustic/CMB baryon density
η_{10}^{ac}	6.170380167710	Derived comparison coordinate	Acoustic baryon-to-photon value
$\Omega_b h_{\text{BBN}}^2$	0.021898765370	Derived comparison coordinate	Transported BBN baryon density
η_{10}^{BBN}	$\eta_{10}^{\text{ac}} \sqrt{\mathcal{G}_T} = 5.998071834744$	Derived comparison coordinate	BBN baryon-to-photon value

Quantity	Relation or numerical value	Class	Role
$N_{\text{eff}}^{\text{CPTG}}$	$p_{\text{ac}} \simeq 2.968584073464$	Derived comparison coordinate	Radiation coordinate
$\Theta_{\text{BBN}}^{\text{CPTG}}$	$(\eta_{10}^{\text{BBN}}, N_{\text{eff}}^{\text{CPTG}}, \tau_7)$	Composite comparison tuple	Primordial-network input tuple; entries retain their individual classes
τ_7	$\ln \pi \simeq 1.1447298858494002$	Abundance-transport comparison coordinate	Live mass-seven optical depth
$\mathcal{S}_7$	$e^{-\tau_7} = 1/\pi \simeq 0.318309886184$	Abundance-transport comparison coordinate	Isolated mass-seven survival
$\mathcal{C}_{r,\mathcal{N}}^{(1)}$	$\text{P}_3 \text{L}_{\mathcal{N}}^{(1)}(\eta_{10}) \mathcal{A}_{r,\mathcal{N}}^{(1)} \mathbf{s}_r$	Network-conditioned comparison coordinate	First-order shared nuclear endpoint response
$\mathcal{D}_c$	0.9735353159758136	Derived comparison coordinate	Cold-sector CMB transport
$\Omega_{c,\text{CPTG}} h^2$	0.120885025627	Frozen comparison input	Pre-map cold-sector input
$\Omega_{c,\text{eff}} h^2$	0.117685841620526	Derived comparison coordinate	CMB cold density
$A_{s,\text{CPTG}}$	2.04×10^{-9}	Frozen comparison input	Pre-map scalar amplitude
$\mathcal{A}_s$	$\pi/3$	Derived comparison coordinate	Scalar-amplitude gate
$A_{s,\text{eff}}$	$2.136283004441 \times 10^{-9}$	Derived comparison coordinate	CMB scalar amplitude
$\tau_{\text{reio},\text{CPTG}}$	0.066	Frozen comparison input	Pre-map reionization optical depth
$\mathcal{T}_\tau$	$\mathcal{G}_T^2$	Derived comparison coordinate	Visibility gate
$\tau_{\text{reio},\text{eff}}$	0.05893087593426084	Derived comparison coordinate	CMB reionization optical depth
n_s	$1 - \pi/100 \simeq 0.968584073464$	Derived comparison coordinate	Scalar tilt
$Y_{\text{He,CMB}}$	0.2485	Frozen comparison input	CMB helium setting
A_{lens}	1	Fixed comparison setting	No phenomenological lensing inflation
$\mathcal{E}_\ell^M$	$(C_\ell^M / C_\ell^{\text{obs}})^{1/2}$	Data-conditioned comparison coordinate	Phase-locked CMB map-envelope transport

Quantity	Defining relation or value	Class	Role
$S_{12}^{\text{CPTG}}(0)$	$K_{S0} I_{C7} = 0.2145003969219307 \text{ eV b}$	Derived comparison coordinate	Zero-energy reaction strength
$S_{12}^{\text{CPTG}}(E)$	$S_{12}^{\text{CPTG}}(0) F_{\text{ext}}(E)$	Data-conditioned comparison coordinate	Declared energy-shape projection
$\mathcal{R}_{pD}^{\text{CPTG}}(T)$	Maxwellian average of $S_{12}^{\text{CPTG}}(E)$	Data-conditioned comparison coordinate	Thermonuclear-rate projection

Table 7: Commissioning-channel reaction-observable coordinates. The zero-energy normalization is fixed by the CPTG commissioning construction. The finite-energy $S(E)$ and thermal-rate coordinates retain the declared external energy-shape carrier and are therefore data-conditioned rather than CPTG-native energy-shape invariants.

Quantity	Relation or value	Class	Role
A_L^{CPTG}	$(p_C/p_{\text{ac}})^2 \simeq 1.119956202138$	Derived branch quantity	Branch-derived lensing response scale
$C_T(a, k)$	$\Phi_\pi(a, k)/\Phi_0(k)$	Bridge-validated comparison coordinate	Curvature-transport amplitude response
$\mathcal{R}_T(a, k)$	$P_\pi(a, k)/P_0(a, k) = C_T(a, k)^2$	Bridge-validated comparison coordinate	Locked power-spectrum response map
$q_{\text{bandpower}}$	$(1 - \mathcal{G}_T)/\mathcal{B}_\pi$	Registered comparison prescription	CMB-lensing bandpower projection weight
$A_{\text{bandpower}}$	1.009690662776	Derived comparison coordinate	Raw CMB-lensing projection
$S_{8, \text{anchor}}$	0.834	Frozen comparison input	Planck-derived amplitude normalization used by the locked late-lensing transport
$S_{8, \text{late}}^{\text{CPTG}}$	$\mathcal{G}_T S_{8, \text{anchor}} = 0.788071335232$	Transported comparison coordinate	Late galaxy-lensing comparison
α_C	$1 - 2/\pi$	Registered comparison prescription	CMB-lensing compression exponent
$S_{8, \text{CMBL}}^{\text{CPTG}}$	0.821191065949	Derived comparison coordinate	CMB-lensing compression
$q_\perp, q_\parallel$	AP distance/ruler ratios	Derived comparison coordinates	DESI/BAO
$q_{\text{iso}}, q_{\text{ap}}$	$(q_\perp^2 q_\parallel)^{1/3}, q_\parallel/q_\perp$	Derived comparison coordinates	ShapeFit
d_f	$f_{\text{CPTG}}/f_{\text{fid}}$	Derived comparison coordinate	ShapeFit growth
d_m	0 in the declared adapter	Fixed comparison setting	ShapeFit tilt placeholder
r_d^{ratio}	$\mathcal{G}_T^{-1/4} \simeq 1.014261942163$	Diagnostic / open derivation	Quarter-ruler candidate
r_d	148.998491 Mpc	Frozen comparison input	Fitted absolute BAO normalization

Table 8: Growth, lensing, curvature-transport spectrum, ShapeFit, and ruler coordinates. The Route B pair C_T and $\mathcal{R}_T = C_T^2$ is bridge-validated at the comparison-coordinate level; this status does not claim a completed native perturbation/source implementation. The late galaxy-lensing amplitude is a transported comparison coordinate: the frozen amplitude normalization is fixed before comparison and the CPTG map is carried by $\mathcal{G}_T$. The quarter-ruler relation remains a supported diagnostic candidate rather than a completed independent derivation.

The numerical inputs and fixed protocol settings used by downstream comparisons have the following upstream provenance. Where the present article carries a value from the locked comparison construction rather than deriving it analytically, that limitation is stated explicitly.

Quantity / setting	Upstream provenance	Freeze rule	Claim boundary / overlap
$\Omega_b h_{\text{ac}}^2 = 0.0225278574944$	Locked CPTG acoustic/CMB comparison-row construction and reproducibility materials [47]	Fixed before CMB and primordial-network evaluations; no sector-specific retuning	Frozen comparison input; not promoted to an analytic native invariant
$\Omega_{\text{e}, \text{CPTG}} h^2 = 0.120885025627$	Locked pre-map CPTG CMB comparison row [47]	Fixed before every cited CMB likelihood evaluation	Frozen comparison input; CPTG subscript denotes row assignment, not native derivation
$A_{\text{s}, \text{CPTG}} = 2.04 \times 10^{-9}$	Locked pre-map CPTG CMB comparison row [47]	Fixed before every cited CMB likelihood evaluation	Frozen comparison input; the derived quantity is $A_{\text{s}, \text{eff}}$
$\tau_{\text{reio}, \text{CPTG}} = 0.066$	Locked pre-map CPTG CMB comparison row [47]	Fixed before low- ℓ , high- ℓ , and combined-stack tests	Frozen comparison input; the derived quantity is $\tau_{\text{reio}, \text{eff}}$
$Y_{\text{He}, \text{CMB}} = 0.2485$	Declared CMB implementation setting	Fixed before likelihood evaluation	Frozen comparison input; not a native helium prediction
$A_{\text{lens}} = 1$	Physical-consistency setting of the CMB comparison protocol [3]	Imposed in every cited CMB evaluation	Fixed comparison setting; distinct from branch-derived A_L^{CPTG}
$S_{8, \text{anchor}} = 0.834$	Planck-derived amplitude normalization [2]	Frozen before DES, HSC, and KiDS late-lensing comparisons	Transport anchor; CMB-data overlap is declared where a CMB-based downstream comparison is used
$d_m = 0$	Declared ShapeFit adapter choice	Fixed throughout the compressed adapter	Fixed comparison setting; no CPTG spectral-tilt prediction is claimed
$r_d = 148.998491 \text{ Mpc}$	Fitted absolute normalization of the existing compressed BAO implementation	Fixed for that compressed BAO calculation; not reclassified by the quarter-ruler candidate	Frozen comparison input; no native absolute-ruler derivation is claimed

Table 9: Provenance and freeze rules for the paper’s frozen comparison inputs and fixed comparison settings.

The symbols $\Omega_{c,\text{CPTG}}h^2$, $A_{s,\text{CPTG}}$, and $\tau_{\text{reio},\text{CPTG}}$ retain the CPTG subscript because they belong to the CPTG pre-map comparison row. The subscript does not change their registry class: they remain frozen comparison inputs, whereas $\Omega_{c,\text{eff}}h^2$, $A_{s,\text{eff}}$, and $\tau_{\text{reio},\text{eff}}$ are the derived comparison coordinates produced by the stated gates.

Likelihood values, χ^2 values, posterior diagnostics, mask row counts, and scan optima are *pipeline results*; they are intentionally excluded from the locked-parameter registry and are reported only in the validation sections together with their method and provenance.

5.1 No-Retuning Rule for Comparison Coordinates

A comparison-layer calculation should follow four steps:

$$\begin{array}{c}
 \text{fix the CPTG-native branch} \\
 \downarrow \\
 \text{choose the observational coordinate system} \\
 \downarrow \\
 \text{apply the locked transport map} \\
 \downarrow \\
 \text{compare without sector-by-sector retuning.}
 \end{array} \tag{33}$$

This rule prevents the comparison layer from becoming a hidden fitting layer. If a parameter is allowed to move during a likelihood calculation, it must be identified as a nuisance or control parameter of the observational pipeline, not as a new CPTG branch parameter.

6 Microscopic Nuclear and Primordial BBN Comparison Layer

The CPTG baryon fraction is fixed by

$$f_b = \frac{\pi}{20} \simeq 0.157079632679. \tag{34}$$

The acoustic/CMB physical baryon-density comparison input is

$$\Omega_b h_{\text{ac}}^2 = 0.0225278574944. \tag{35}$$

The analysis uses the standard fixed-temperature conversion convention [13]

$$\eta_{10} \simeq 273.9 \Omega_b h^2, \tag{36}$$

so that the stored-precision acoustic baryon-density coordinate gives

$$\eta_{10}^{\text{ac}} = 6.170380167710. \tag{37}$$

The displayed $\Omega_b h_{\text{ac}}^2$ value is rounded relative to the stored numerical value used by the conversion, so reconstruction from the printed decimal can differ only in the last displayed digits.

The effective-radiation comparison coordinate is identified with the acoustic exponent,

$$N_{\text{eff}}^{\text{CPTG}} = p_{\text{ac}} \simeq 2.968584073464. \quad (38)$$

For light-element comparison, the baryon coordinate transports as

$$\eta_{10}^{\text{BBN}} = \eta_{10}^{\text{ac}} \sqrt{\mathcal{G}_T} = 5.998071834744, \quad (39)$$

$$\Omega_b h_{\text{BBN}}^2 = \Omega_b h_{\text{ac}}^2 \sqrt{\mathcal{G}_T} = 0.021898765370. \quad (40)$$

Thus

$$(\Omega_b h_{\text{ac}}^2, \eta_{10}^{\text{ac}}) \longrightarrow (\Omega_b h_{\text{BBN}}^2, \eta_{10}^{\text{BBN}}) \quad (41)$$

is a comparison-coordinate transport rather than a refit of primordial abundances.

For reference, the Standard Model radiation benchmark used by the modern BBN literature is $N_{\text{eff}}^{\text{SM}} = 3.044$ [25], while Planck combined with BAO constrains the corresponding observational extension to $N_{\text{eff}} = 2.99 \pm 0.17$ [2]. These are different objects: the former is a theoretical Standard Model benchmark and the latter is an observational constraint. They are therefore not combined into a hybrid “central value plus error.”

6.1 Commissioning-Channel Reaction-Observable Coordinates

The commissioned $D(p, \gamma)^3\text{He}$ calculation supplies a direct bridge from a reaction-specific CPTG source construction into conventional nuclear-astrophysics observables. Its ordering is

$$\begin{aligned} & \text{reaction source amplitude} \\ & \quad \downarrow \\ & \text{dimensionless source intensity} \\ & \quad \downarrow \\ & S_{12}^{\text{CPTG}}(0) \\ & \quad \downarrow \\ & S_{12}^{\text{CPTG}}(E) \\ & \quad \downarrow \\ & \mathcal{R}_{pD}^{\text{CPTG}}(T). \end{aligned} \quad (42)$$

The retained coherent commissioning amplitude is $\mathcal{A}_{C6} = 17.5595746398194$, giving the dimensionless native source intensity

$$I_{C7} = |\mathcal{A}_{C6}|^2 = 308.3386615313886. \quad (43)$$

The amplitude contraction itself is reaction-specific and belongs to the commissioning construction; the comparison-coordinate step begins when this native dimensionless intensity is converted into an observable with units [15].

For the zero-energy reaction-strength coordinate, the dimensional bridge is

$$K_{S0} = \frac{\alpha\pi}{33 - 1/(7\pi)} \text{ eV b} = 0.0006956649414530029 \text{ eV b}, \quad (44)$$

so that

$$S_{12}^{\text{CPTG}}(0) \equiv K_{S0} I_{C7} = 0.2145003969219307 \text{ eV b}. \quad (45)$$

The support-count denominator in K_{S0} is fixed by the commissioned $A = 3$ construction before external S -factor comparison; it is not fitted to the LUNA extrapolation. The corresponding observational checks are recorded later in Section 12.1.

The finite-energy coordinate requires a stricter classification. The commissioned construction fixes the zero-energy normalization, but its low-energy shape is supplied by a declared external carrier. Write

$$S_{12}^{\text{CPTG}}(E) = S_{12}^{\text{CPTG}}(0) F_{\text{ext}}(E). \quad (46)$$

For the Solar Fusion II low-energy carrier used in the commissioning analysis,

$$F_{\text{SFII}}(E) = \frac{S_{12,\text{SFII}}(E)}{S_{12,\text{SFII}}(0)}, \quad (47)$$

with

$$S_{12,\text{SFII}}(E) = \left[0.2145 + 5.5612 \left(\frac{E}{\text{MeV}} \right) + 4.6581 \left(\frac{E}{\text{MeV}} \right)^2 \right] \text{ eV b}. \quad (48)$$

At $E_0 = 6.64 \text{ keV}$, $F_{\text{SFII}} = 1.1731083532203264$ and

$$S_{12}^{\text{CPTG}}(E_0) = 0.2516322073981925 \text{ eV b}, \quad (49)$$

The energy-shape function is part of the declared comparison coordinate and is not relabeled as a new CPTG-native invariant. Its observational comparison is recorded in Section 12.1.

For charged-particle capture,

$$\sigma(E) = \frac{S(E)}{E} \exp[-2\pi\eta(E)], \quad \eta(E) = Z_1 Z_2 \alpha \sqrt{\frac{\mu c^2}{2E}}, \quad (50)$$

with $Z_1 = Z_2 = 1$ and $\mu = m_p m_D / (m_p + m_D)$ for this channel. The thermonuclear comparison coordinate is the Maxwellian pushforward

$$\mathcal{R}_{pD}^{\text{CPTG}}(T) \equiv N_A \left(\frac{8}{\pi\mu} \right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty S_{12}^{\text{CPTG}}(E) \exp\left[-\frac{E}{k_B T} - 2\pi\eta(E)\right] dE. \quad (51)$$

Because this thermal coordinate inherits the declared $S(E)$ shape carrier, it is data-conditioned at finite energy even though its CPTG zero-energy normalization is fixed. A tabulated network may implement the same normalization through

$$R_{pD}^{\text{CPTG}}(T) = \lambda_{pD} R_{pD}^{\text{table}}(T), \quad \lambda_{pD} = 1.0000018504518913, \quad (52)$$

where $\lambda_{pD} = S_{12}^{\text{CPTG}}(0)/S_{12}^{\text{table}}(0)$ with $S_{12}^{\text{table}}(0) = 0.2145000000000000 \text{ eV b}$. This ratio is an implementation adapter, not a new geometric parameter [15].

The completed commissioning-channel observational checks are collected in Section 12.1; they do not modify the construction above.

6.2 Network-Conditioned Nuclear Reaction-Response Pushforward

The primordial input tuple above specifies the background and the separate live mass-seven transport. The deeper reaction-response comparison begins from the universal four-sector source coordinate rather than from an endpoint abundance vector. For reaction r , baryon and charge conservation leave three ordered transport coordinates and one independent polarization coordinate,

$$3\delta P_V + \delta P_C = 0, \quad \Pi = \delta P_V = -\frac{\delta P_C}{3}, \quad (53)$$

so that the exact reaction-source coordinate is

$$\mathbf{s}_r = \begin{pmatrix} q_{VB} \\ q_{BC} \\ q_{CS} \\ \Pi \end{pmatrix}_r. \quad (54)$$

Reaction stoichiometry fixes this source direction before numerical endpoint responses are opened; it is not a fitted abundance direction [22].

A native network $\mathcal{N}$ supplies its own code-local source kernel. At first order in the logarithmic rate coordinate $x = -\ln f_r$,

$$\mathcal{F}_{r,\mathcal{N}}(x) = \mathcal{A}_{r,\mathcal{N}}^{(1)}x + \mathcal{O}(x^2), \quad (55)$$

which generates the source drive

$$\mathbf{d}_{r,\mathcal{N}}^{(1)} = \mathcal{A}_{r,\mathcal{N}}^{(1)}\mathbf{s}_r. \quad (56)$$

The native network dynamics then define the observable pushforward

$$\mathbf{y}_{r,\mathcal{N}}^{(1)} = \mathbf{L}_{\mathcal{N}}^{(1)}(\eta_{10})\mathbf{d}_{r,\mathcal{N}}^{(1)}. \quad (57)$$

Neither the numerical entries nor even the declared observable dimension of $\mathbf{L}_{\mathcal{N}}^{(1)}$ are assumed to be identical between PRIMAT, PArthENoPE, and PRyMordial. The universal object is the fixed source geometry and factorization; the numerical pushforward remains network-local [22].

For the shared primordial endpoint comparison used in the present authority chain, define

$$\mathbf{P}_3\mathbf{y} = \begin{pmatrix} \Delta \ln(D/H) \\ \Delta \ln(^3\text{He}/H) \\ \Delta \ln(^7\text{Li}/H) \end{pmatrix}. \quad (58)$$

The first-order network-conditioned nuclear comparison coordinate is therefore

$$\mathcal{C}_{r,\mathcal{N}}^{(1)}(\eta_{10}) \equiv \mathbf{P}_3\mathbf{y}_{r,\mathcal{N}}^{(1)} = \mathbf{P}_3\mathbf{L}_{\mathcal{N}}^{(1)}(\eta_{10})\mathcal{A}_{r,\mathcal{N}}^{(1)}\mathbf{s}_r. \quad (59)$$

This is a comparison coordinate, not a new native invariant. It depends on the fixed CPTG source direction, the declared native network, the baryon-density coordinate, and the chosen observable projection. The bare source vector $\mathbf{s}_r$ is therefore not identified directly with the final abundance-response vector. The completed PRIMAT source/Jacobian campaign validates the dynamical pushforward itself, rather than a direct source-to-endpoint identification [22, 14].

6.3 Primordial Nuclear and Mass-Seven Abundance-Transport Coordinates

The comparison layer supplied to a native primordial network is conveniently collected as

$$\Theta_{\text{BBN}}^{\text{CPTG}} = (\eta_{10}^{\text{BBN}}, N_{\text{eff}}^{\text{CPTG}}, \tau_7). \quad (60)$$

The first two entries are the transported baryon and radiation coordinates above. The third is a distinct live abundance-transport coordinate. In the geometric pi-branch comparison construction, the integrated mass-seven transport depth is fixed as

$$\tau_7 \equiv \int \Gamma_{7,\text{trans}}(t) dt = \ln \pi \simeq 1.1447298858494002. \quad (61)$$

The corresponding isolated-channel survival coordinate is therefore

$$\mathcal{S}_7 \equiv e^{-\tau_7} = \frac{1}{\pi} \simeq 0.3183098861837907. \quad (62)$$

The pi-branch comparison construction fixes $\tau_7 = \ln \pi$ before the downstream nuclear-reaction and lithium validations. The downstream nuclear-reaction and cosmological-lithium analyses apply and test that already-fixed law rather than deriving it from the lithium endpoint [15, 22, 23].

The live network action is not a post-processing division of a completed abundance. For the two surviving mass-seven channels it has the comparison-level form

$$\frac{dY_i}{dt} \longrightarrow \left(\frac{dY_i}{dt} \right)_{\text{BBN}} - \Gamma_{7,\text{trans}}(t)Y_i, \quad i \in \{^7\text{Li}, ^7\text{Be}\}. \quad (63)$$

The exact relation $\mathcal{S}_7 = 1/\pi$ is the isolated survival law. A live native network continues ordinary production, destruction, and redistribution while the transport term is active, so the measured gate-off/gate-on endpoint ratio is a network-response quantity and is tested against, rather than algebraically forced to equal, $1/\pi$.

The four-sector CPTG nuclear source geometry remains a separate parent structure: free nucleons form the vertex, deuterium the bridge, tritium/helium-3 the closure sector, and helium-4 the saturation sector. Mass seven is not promoted to a fifth baryon-simplex coordinate. The $A = 7$ law is a distinct downstream abundance-transport action on the evolving ^7Li and ^7Be variables, not an asserted conventional nuclear-reaction channel [22].

7 Locked CMB Acoustic Transport Layer

The CPTG acoustic/CMB projection is fixed before likelihood evaluation. The native quantity p_C , the derived acoustic quantities p_{ac} , $\mathcal{G}_T$, and $\sqrt{\mathcal{G}_T}$, together with the CMB inputs and output coordinates, are

registered in Tables 4, 6, and 9. The active native-to-comparison map is

$$\Omega_{c,\text{eff}} h^2 = \Omega_{c,\text{CPTG}} h^2 \mathcal{D}_c, \quad (64)$$

$$A_{s,\text{eff}} = A_{s,\text{CPTG}} \mathcal{A}_s, \quad (65)$$

$$\tau_{\text{reio},\text{eff}} = \tau_{\text{reio},\text{CPTG}} \mathcal{T}_\tau. \quad (66)$$

The cold-sector transport is

$$\mathcal{D}_c = \sqrt{\mathcal{G}_T} \left(1 + \frac{\pi - 3}{30\pi} \right) = 0.9735353159758136, \quad (67)$$

while the amplitude and visibility gates are

$$\mathcal{A}_s = \frac{\pi}{3} = 1.0471975511965976, \quad \mathcal{T}_\tau = \mathcal{G}_T^2 = 0.8928920596100127. \quad (68)$$

Their geometric origin is derived in Section 7.1; their arithmetic closure against the fixed row is tested in Section 13.10.

Parameter	Locked comparison value	Registry class
H_0	67.4777967351 km s ⁻¹ Mpc ⁻¹	Derived comparison coordinate
$\Omega_b h^2$	0.022527857494	Frozen comparison input
$\Omega_c h^2$	0.117685841620526	Derived comparison coordinate
n_s	0.968584073464	Derived comparison coordinate
N_{eff}	2.968584073464	Derived comparison coordinate
$Y_{\text{He,CMB}}$	0.2485	Frozen comparison input
A_s	$2.136283004441 \times 10^{-9}$	Derived comparison coordinate
τ_{reio}	0.058930875934	Derived comparison coordinate
A_{lens}	1	Fixed comparison setting

Table 10: Locked CPTG CMB acoustic-transport comparison row.

For arithmetic-precision bookkeeping, the corresponding unrounded mapped values are $\Omega_{c,\text{eff}} h^2 = 0.11768584162052577$, $A_{s,\text{eff}} = 2.1362830044410592 \times 10^{-9}$, and $\tau_{\text{reio},\text{eff}} = 0.05893087593426084$. The table retains the declared comparison-row display precision.

This row is fixed before individual likelihood evaluations and is not retuned separately for Plik, CamSpec, SPT, low- ℓ , or lensing channels. The likelihood setting $A_{\text{lens}} = 1$ is distinct from the branch-derived lensing response scale A_L^{CPTG} ; no phenomenological $A_{\text{lens}} > 1$ enhancement is introduced [3, 2].

7.1 Geometric Origin of the CMB Amplitude and Visibility Gates

The locked CMB comparison layer contains two geometric projection gates beyond the cold-sector distance map. They are derived from the registered branch quantities rather than fitted to CMB residuals.

7.1.1 Amplitude Projection and the $\pi/3$ Curvature Gate

The scalar amplitude comparison is a three-spatial-direction projection of the registered curvature branch $p_C = \pi$. The branch-to-acoustic projection is therefore

$$\mathcal{A}_C = \frac{p_C}{3} = \frac{\pi}{3} = 1.0471975511965976. \quad (69)$$

The comparison map uses $\mathcal{A}_s \equiv \mathcal{A}_C$, so

$$A_{s,\text{eff}} = A_{s,\text{CPTG}} \frac{\pi}{3}. \quad (70)$$

The frozen value $A_{s,\text{CPTG}} = 2.04 \times 10^{-9}$ is an input to this map, not an independently derived native amplitude; its provenance and freeze rule are given in Table 9.

7.1.2 Visibility Transport and the $\mathcal{G}_T^2$ Reionization-Depth Gate

The reionization-depth comparison coordinate is carried by the second-order acoustic transport visibility projection

$$\mathcal{T}_\tau = \mathcal{G}_T^2 = 0.8928920596100127. \quad (71)$$

The resulting map is

$$\tau_{\text{reio},\text{eff}} = \tau_{\text{reio},\text{CPTG}} \mathcal{G}_T^2. \quad (72)$$

The frozen value $\tau_{\text{reio},\text{CPTG}} = 0.066$ is a pre-map comparison input and is not retuned against low- ℓ polarization. The independent numerical bookkeeping test of both gates, including the effective values $A_{s,\text{eff}} = 2.136283004441059 \times 10^{-9}$ and $\tau_{\text{reio},\text{eff}} = 0.05893087593426084$, is reported in Section 13.10.

8 CMB Comparison Coordinates and Reference Quantities

The locked CPTG CMB row is a fixed geometric construction expressed in the coordinates used by CMB likelihoods. External reference quantities do not all have the same provenance: some are Planck posterior values, some are theoretical Standard Model benchmarks, and some are BBN-consistency-derived or convention-converted quantities. Keeping those classes separate prevents a theoretical benchmark from being misread as a direct posterior measurement.

Quantity	CPTG comparison value	External reference	Reference class
H_0	67.4777967351	67.36	Planck base- Λ CDM posterior [2]
$\Omega_b h^2$	0.022527857494	0.02237	Planck base- Λ CDM posterior [2]
$\Omega_c h^2$	0.117685841621	0.1200	Planck base- Λ CDM posterior [2]
n_s	0.968584073464	0.9649	Planck base- Λ CDM posterior [2]
N_{eff}	2.968584073464	3.044	Standard Model BBN benchmark [25]
N_{eff}	2.968584073464	2.99 ± 0.17	Planck+BAO one-parameter extension [2]
$Y_{\text{He,CMB}}$	0.2485	$\simeq 0.2454$	Planck BBN-consistency-derived helium quantity [2]
A_s	$2.136283004441 \times 10^{-9}$	2.100549×10^{-9}	Converted Planck scalar-amplitude convention [2]
τ_{reio}	0.058930875934	0.0544	Planck reionization-optical-depth posterior [2]

Table 11: CPTG CMB comparison coordinates and external references, separated by provenance class.

The $N_{\text{eff}} = 3.044$ row is a theoretical Standard Model benchmark, while 2.99 ± 0.17 is an observational Planck+BAO constraint in an extended parameter space. They are not combined into a single “measured” reference value. Likewise, the Planck Y_{He} entry is BBN-consistency-derived rather than an independent direct CMB helium measurement.

The role of the external row is to provide reference scale and provenance. The actual fixed-row decision is made from like-for-like likelihood shifts,

$$\Delta\chi^2 = \chi_{\text{CPTG,locked}}^2 - \chi_{\text{control}}^2, \quad (73)$$

under identical likelihood and nuisance machinery. The CPTG coordinates therefore need not numerically equal each external central value one by one in order for the fixed row to remain at control level.

9 Recombination-Scale Perturbation and CMB Projection Construction

CPTG treats growth and lensing as reduced perturbation channels of the nonlinear curvature-polarization and curvature-transport field. To interface with conventional CMB and lensing likelihoods, this section uses the scalar metric, effective Poisson relation, and Weyl-potential combination below solely as external comparison-coordinate definitions. They are not asserted as CPTG-native field equations, do not make GR or FLRW the governing CPTG framework, and do not introduce a dark-matter sector. The functions μ_{CPTG} and Σ_{CPTG} are translation objects at this boundary. In this comparison coordinate, the scalar metric is written as

$$ds^2 = -(1 + 2\Psi) c^2 dt^2 + a(t)^2 (1 - 2\Phi) d\mathbf{x}^2. \quad (74)$$

The growth response is defined by the effective Poisson relation

$$k^2 \Psi = -4\pi G a^2 \bar{\rho} \mu_{\text{CPTG}}(a, k) \delta. \quad (75)$$

The lensing response is defined through the Weyl-potential combination

$$k^2 (\Phi + \Psi) = -8\pi G a^2 \bar{\rho} \Sigma_{\text{CPTG}}(a, k) \delta. \quad (76)$$

9.1 Curvature-Transport Spectrum Bridge

The locked Route B Option 1 comparison coordinate supplies an amplitude-level curvature-transport bridge between the CPTG potential response and a conventional spectrum coordinate. The amplitude comparison coordinate is

$$C_T(a, k) \equiv \frac{\Phi_\pi(a, k)}{\Phi_0(k)}, \quad (77)$$

so that

$$\Phi_\pi(a, k) = \Phi_0(k) C_T(a, k). \quad (78)$$

Because a power spectrum is quadratic in the corresponding amplitude, the locked power-response comparison coordinate is

$$\mathcal{R}_T(a, k) \equiv \frac{P_\pi(a, k)}{P_0(a, k)} = C_T(a, k)^2. \quad (79)$$

The identity limit is therefore

$$C_T(a, k) = 1 \implies \mathcal{R}_T(a, k) = 1 \implies P_\pi(a, k) = P_0(a, k). \quad (80)$$

Route B Option 1 locks the squared-amplitude response as the primary comparison mapping. Its validation boundary is deliberately narrower than a native perturbation-theory claim: it is a geometry-first bridge validation, not a completed CPTG Boltzmann/source implementation, and it uses the squared-amplitude response rather than an f -to- A_s proxy as its physical map [41]. The bridge lock fixes the amplitude-to-power response relation; it does not by itself supply an independent closed-form native evolution law for $C_T(a, k)$.

The implementation record for this bridge is given in Section 13.1; those stress metrics do not redefine C_T or $\mathcal{R}_T$.

9.2 Data-Conditioned CMB Map-Envelope Coordinate

The real-map comparison layer uses the locked CMB angular-power envelope without claiming that CPTG independently predicts the random CMB phase realization. For an observed temperature field with harmonic coefficients $a_{\ell m}^{\text{obs}}$ and observed angular spectrum C_ℓ^{obs} , define the data-conditioned envelope coordinate for model M by

$$\mathcal{E}_\ell^M \equiv \left(\frac{C_\ell^M}{C_\ell^{\text{obs}}} \right)^{1/2}, \quad \ell \geq 2. \quad (81)$$

The corresponding phase-locked harmonic comparison coefficients are

$$a_{\ell m}^M = a_{\ell m}^{\text{obs}} \mathcal{E}_\ell^M, \quad (82)$$

and the real-space comparison map is

$$T_M(\hat{n}) = \sum_{\ell m} a_{\ell m}^M Y_{\ell m}(\hat{n}). \quad (83)$$

This construction makes the observational conditioning explicit: the phase scaffold comes from the observed sky, while the CPTG or control model supplies the angular-power envelope. The data scaffold is therefore part of the comparison coordinate, not an additional CPTG-native parameter [42].

The declared map-space projection permits only a global amplitude and offset,

$$T_{M,\text{fit}}(\hat{n}) = A_M \tilde{T}_M(\hat{n}) + B_M, \quad (84)$$

with residual fraction

$$R_M \equiv \frac{\text{RMS}[T_{\text{obs}}(\hat{n}) - T_{M,\text{fit}}(\hat{n})]}{\text{RMS}[T_{\text{obs}}(\hat{n})]}, \quad (85)$$

and relative comparison gap

$$\Delta R \equiv R_{\text{CPTG}} - R_{\text{Planck}}. \quad (86)$$

Here A_M , B_M , R_M , and ΔR are observational projection or validation quantities; they are not promoted into the CPTG theory registry.

The implementation and null-envelope results for this coordinate are recorded in Section 13.2; they do not alter the map definition above.

The construction boundary remains narrow. This is a real-map comparison-coordinate closure test using a declared observed phase scaffold. It is not a native prediction of the exact random CMB phases, and the small Planck-over-CPTG edge is not assigned uniquely to theory because smoothing, mask, sky-selection, component-separation, beam/transfer, and comparison-coordinate controls all contribute to its attribution budget [42].

10 Late-Time Growth and Lensing Construction

The perturbation definitions introduced at the recombination-scale transition are carried forward into the post-recombination structure-growth regime. The following quantities are therefore organized here by their late-time transport role rather than by the observational instrument that measures them.

For the reduced comparison layer used here, the active scale-averaged growth and lensing responses are written as fixed compressed-response forms built from the branch-derived A_L^{CPTG} response scale. These relations organize the comparison calculation; they are not presented here as a completed native perturbation derivation:

$$\mu_{\text{CPTG}}(a) = 1 + \frac{\pi}{2} \left(A_L^{\text{CPTG}} - 1 \right) (1 - a^{p_{\text{ac}}}), \quad (87)$$

$$\Sigma_{\text{CPTG}}(a) = 1 + \left(A_L^{\text{CPTG}} - 1 \right) (1 - a^{p_{\text{ac}}}). \quad (88)$$

The branch-derived lensing response scale is fixed by the curvature-to-acoustic exponent ratio:

$$A_L^{\text{CPTG}} = \left(\frac{p_C}{p_{\text{ac}}} \right)^2. \quad (89)$$

With

$$p_C = \pi \quad (90)$$

and

$$p_{\text{ac}} = 3 - \frac{\pi}{100}, \quad (91)$$

this gives

$$A_L^{\text{CPTG}} = 1.119956202138. \quad (92)$$

The raw CMB-lensing bandpower comparison is not the same object as the native lensing response. Its registered projection prescription gives the bandpower amplitude

$$A_{\text{bandpower}} = 1 + q_{\text{bandpower}} \left(A_L^{\text{CPTG}} - 1 \right), \quad (93)$$

where

$$q_{\text{bandpower}} = \frac{1 - \mathcal{G}_T}{\mathcal{B}_\pi}. \quad (94)$$

Therefore,

$$A_{\text{bandpower}} = 1.009690662776. \quad (95)$$

The late-time galaxy-lensing amplitude is a transported comparison coordinate rather than an additional native invariant. The locked construction uses the frozen Planck-derived comparison normalization

$$S_{8,\text{anchor}} = 0.834,$$

and applies the CPTG geometric transport factor,

$$S_{8,\text{late}}^{\text{CPTG}} = \mathcal{G}_T S_{8,\text{anchor}} = 0.788071335232. \quad (96)$$

The anchor is fixed before the late-lensing comparison and is not re-estimated against DES, HSC, or KiDS data [2].

The compressed CMB-lensing amplitude channel is

$$S_{8,\text{CMBL}}^{\text{CPTG}} = S_{8,\text{late}}^{\text{CPTG}} \left(A_L^{\text{CPTG}} \right)^{\alpha_C}. \quad (97)$$

The registered compression exponent is

$$\alpha_C = 1 - \frac{2}{\pi}. \quad (98)$$

This gives

$$S_{8,\text{CMBL}}^{\text{CPTG}} = 0.821191065949. \quad (99)$$

Thus CPTG distinguishes three lensing-related quantities:

$$A_L^{\text{CPTG}}, \quad A_{\text{bandpower}}, \quad S_{8,\text{CMBL}}^{\text{CPTG}}. \quad (100)$$

The first is the branch-derived response scale, the second is a projected raw bandpower comparison, and the third is a compressed amplitude comparison. These are different observables and should not be collapsed into a single fitted lensing parameter.

The observational weak-lensing and CMB-lensing comparisons for these coordinates are collected in the late-time verification record, Section 15.1.

11 Large-Scale Distance, Horizon, and Acoustic-Ruler Comparison Layer

Before observational conversion, the relevant background relations are fixed by the native π branch. The late and transport weights are

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}, \quad (101)$$

and the native expansion function and its exact acoustic projection are

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \quad (102)$$

$$E_{\text{ac}}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}. \quad (103)$$

Equivalently, E_{ac} is generated from E_π by the projector defined in the core conventions; it is not a second native background.

The acceleration anchor fixes the asymptotic rate and curvature scale through

$$H_\infty = \frac{a_{\pi,0}}{c} = 57.3131992073 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (104)$$

$$H_0^{(\pi)} = \frac{H_\infty}{\sqrt{\mathcal{B}_\pi}} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (105)$$

$$R_{C0} = 48\pi \frac{c^2}{a_{\pi,0}} = 788.7823091477 \text{ Gpc}. \quad (106)$$

The curvature radius evolves on the native branch according to

$$R_C(a) = \frac{R_{C0}}{E_\pi(a)}, \quad H_\pi(a)R_C(a) = H_0^{(\pi)}R_{C0}, \quad (107)$$

with $H_\pi(a) = H_0^{(\pi)}E_\pi(a)$. This is the curvature-radius evolution used by the companion CPTG cosmological-horizon analysis [40]; it is a native geometric relation and is not a GR/FLRW horizon substitution.

Hubble-like rates are transported by $\sqrt{\mathcal{G}_T}$, giving

$$H_{0,\text{ac}}^{\text{CPTG}} = H_0^{(\pi)}\sqrt{\mathcal{G}_T} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (108)$$

$$H_{\infty,\text{ac}}^{\text{CPTG}} = H_\infty\sqrt{\mathcal{G}_T} = 55.7127237837 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (109)$$

The acoustic comparison weights follow directly from the acoustic exponent:

$$\Omega_{m,\text{ac}} = \frac{p_{\text{ac}}}{3\pi} \simeq 0.314976552850, \quad (110)$$

$$\Omega_{\Lambda,\text{ac}} = 1 - \frac{p_{\text{ac}}}{3\pi} \simeq 0.685023447150. \quad (111)$$

These relations establish a single analytic source for the displayed decimals.

11.1 Acoustic-Branch Clock and Density-Weighted Diagnostic

The acoustic branch is the exact projection of the native π branch defined above. Its self-synchronized clock uses the acoustic projected rate

$$E_{\text{ac}}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad (112)$$

with $H_{\text{ac}}(a) = H_{0,\text{ac}}^{\text{CPTG}}E_{\text{ac}}(a)$ and present age

$$t_{0,\text{ac}}^{\text{CPTG}} = \frac{1}{H_{0,\text{ac}}^{\text{CPTG}}} \int_0^1 \frac{da}{aE_{\text{ac}}(a)} = 13.88517358 \text{ Gyr}. \quad (113)$$

This is the acoustic branch time coordinate, not a universal clock imposed on the native or luminosity branches.

All dimensional ages in this section use $1 \text{ Mpc} = 3.0856775814913673 \times 10^{19} \text{ km}$ and the Julian year $1 \text{ yr} = 31,557,600 \text{ s}$, so $1 \text{ Gyr} = 3.15576 \times 10^{16} \text{ s}$. This convention fixes the dimensional conversion of the age integrals and reproduces the displayed full-precision acoustic-density value below.

The term *acoustic-density* is used here narrowly for a CMB density-weighted diagnostic overlay. The external CMB pipeline carries baryon and conventional cold-sector density coordinates that parameterize its acoustic observational coordinate; those inputs are not promoted into a CPTG dark-matter sector. The diagnostic does not insert those physical densities directly; it uses the dimensionless weights $\Omega_{m,\text{ac}}$ and $\Omega_{\Lambda,\text{ac}}$. These weights are distinct from the exact projector weights $\mathcal{T}_\pi$ and $\mathcal{B}_\pi$. Retaining the established acoustic exponent gives

$$E_{\text{ac},\rho}(a)^2 = \Omega_{\Lambda,\text{ac}} + \Omega_{m,\text{ac}} a^{-p_{\text{ac}}}. \quad (114)$$

Thus “density” identifies only the CMB density weighting of this diagnostic. $E_{\text{ac},\rho}$ is not another acoustic branch, does not replace E_{ac} , and does not define an additional branch clock. Its reported age-like diagnostic is

$$t_{0,\text{ac},\rho}^{\text{CPTG}} = \frac{1}{H_{0,\text{ac}}^{\text{CPTG}}} \int_0^1 \frac{da}{a E_{\text{ac},\rho}(a)} \quad (115)$$

$$= \frac{2}{p_{\text{ac}} H_{0,\text{ac}}^{\text{CPTG}} \sqrt{\Omega_{\Lambda,\text{ac}}}} \sinh^{-1} \sqrt{\frac{\Omega_{\Lambda,\text{ac}}}{\Omega_{m,\text{ac}}}} \quad (116)$$

$$= 13.926455176722689 \text{ Gyr}. \quad (117)$$

At six-decimal display precision this value is 13.926455 Gyr. The same p_{ac} -weighted diagnostic background is independently present in the CPTG compressed growth/RSD implementations, providing an implementation cross-check [47]. It remains a diagnostic overlay on the acoustic observational coordinate and is not promoted into the three-branch cosmological architecture.

11.2 Luminosity Projection, Clock, and Endpoint

The local luminosity coordinate is observationally conditioned rather than independently fixed by the native branch. Using the four-anchor SH0ES distance-ladder value [4],

$$H_0^{\text{SH0ES}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (118)$$

the present-epoch luminosity-rate coordinate is

$$\mathcal{L}_{\text{loc}} = \frac{H_0^{\text{SH0ES}}}{H_0^{(\pi)}} = 1.054075942846, \quad (119)$$

$$\sigma_{\mathcal{L}} = \mathcal{L}_{\text{loc}} \frac{0.86}{73.17} = 0.012389029805. \quad (120)$$

Thus

$$\mathcal{L}_{\text{loc}} = 1.054075942846 \pm 0.012389029805. \quad (121)$$

The reciprocal present-epoch rate factor is

$$\Xi_{\text{loc}}(1) = \mathcal{L}_{\text{loc}}^{-1} = 0.948698247775, \quad (122)$$

$$\sigma_{\Xi} = \Xi_{\text{loc}}(1) \frac{0.86}{73.17} = 0.011150478244. \quad (123)$$

The equivalent luminosity endpoint is therefore

$$H_{0,\text{lum}}^{\text{CPTG}} = H_0^{(\pi)} \mathcal{L}_{\text{loc}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (124)$$

These quantities are data-conditioned comparison coordinates: SH0ES supplies the local observational normalization, while the CPTG native normalization supplies the reference denominator. They must not be cited as an independent native prediction of the SH0ES endpoint.

The luminosity branch is a projection of native time. In general, write

$$H_{\text{lum}}(N_\pi) = \mathcal{L}(N_\pi) H_\pi(N_\pi), \quad \Xi_{\text{lum}}(N_\pi) \equiv \mathcal{L}(N_\pi)^{-1}. \quad (125)$$

The branch clocks then satisfy

$$\frac{dt_{\text{lum}}}{dt_\pi} = \Xi_{\text{lum}}(N_\pi), \quad H_{\text{lum}} dt_{\text{lum}} = H_\pi dt_\pi = dN_\pi. \quad (126)$$

Thus luminosity time is not an endpoint reciprocal or a universal rescaling of native time; it is the integrated time coordinate generated by the phase-dependent luminosity rate.

For the full-history comparison realization used in the CPTG Hubble-bridge and horizon analyses [39, 40], the SH0ES-conditioned luminosity rate is

$$H_{\text{lum}}^{\text{anchor}}(a) = H_{0,\text{lum}}^{\text{CPTG}} E_{\text{ac},\rho}(a), \quad (127)$$

so that

$$\Xi_{\text{lum}}^{\text{anchor}}(a) = \frac{H_\pi(a)}{H_{\text{lum}}^{\text{anchor}}(a)} \quad (128)$$

$$= \frac{H_0^{(\pi)}}{H_{0,\text{lum}}^{\text{CPTG}}} \left[\frac{\mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}}{\Omega_{\Lambda,\text{ac}} + \Omega_{m,\text{ac}} a^{-p_{\text{ac}}}} \right]^{1/2}. \quad (129)$$

This recovers $\Xi_{\text{lum}}^{\text{anchor}}(1) = \Xi_{\text{loc}}(1)$ and gives

$$t_{0,\text{lum}}^{\text{anchor}} = \int_0^1 \frac{da}{a H_{\text{lum}}^{\text{anchor}}(a)} = 12.8430574232 \text{ Gyr}. \quad (130)$$

This is a data-conditioned comparison realization of the luminosity projection, not a claim that $E_{\text{ac},\rho}$ is the microscopic luminosity projector. No additional cosmological branch is introduced.

The acoustic-to-luminosity endpoint separation in this comparison is

$$\frac{H_{0,\text{lum}}^{\text{CPTG}}}{H_{0,\text{ac}}^{\text{CPTG}}} - 1 = 0.0843566853 = 8.43566853\%. \quad (131)$$

11.3 Acoustic–Luminosity Midpoint Comparison Coordinate

Some CPTG sensitivity calculations require a central comparison coordinate between the acoustic and luminosity endpoints without promoting that central value to another cosmological branch. The registered symmetric midpoint prescription is therefore

$$\omega_{\text{AL}} \equiv \frac{1}{2}. \quad (132)$$

For any two endpoint quantities $Q_{\text{ac}}(N_\pi)$ and $Q_{\text{lum}}(N_\pi)$ that have first been evaluated in the same comparison coordinate at the same native phase, the midpoint readout is

$$Q_{\text{mid}}(N_\pi) \equiv (1 - \omega_{\text{AL}})Q_{\text{ac}}(N_\pi) + \omega_{\text{AL}}Q_{\text{lum}}(N_\pi). \quad (133)$$

The weight $\omega_{\text{AL}} = 1/2$ is the CPTG midpoint prescription. It is a registered comparison rule, not an additional native invariant, branch exponent, or fitted parameter. The resulting midpoint value inherits the registry status and observational conditioning of the endpoint quantities to which the rule is applied.

For the rate coordinate, Eq. (133) gives

$$H_{\text{mid}}^{\text{CPTG}}(a) \equiv (1 - \omega_{\text{AL}})H_{\text{ac}}(a) + \omega_{\text{AL}}H_{\text{lum}}^{\text{anchor}}(a) = \frac{1}{2} [H_{\text{ac}}(a) + H_{\text{lum}}^{\text{anchor}}(a)], \quad (134)$$

with dimensionless native-relative rate coordinate

$$\mathcal{M}_{\text{AL}}(a) \equiv \frac{H_{\text{mid}}^{\text{CPTG}}(a)}{H_\pi(a)}. \quad (135)$$

Because both endpoint rates are positive, this rate-space realization lies between the acoustic and luminosity rate coordinates at the same N_π . It is not an additional branch or an interpolation of the underlying native geometry.

At the normalized present phase,

$$\begin{aligned} \mathcal{M}_{\text{AL},0} &= (1 - \omega_{\text{AL}})\sqrt{\mathcal{G}_T} + \omega_{\text{AL}}\mathcal{L}_{\text{loc}} \\ &= 1.013075433598 \pm 0.006194514903, \end{aligned} \quad (136)$$

and therefore

$$\begin{aligned} H_{0,\text{mid}}^{\text{CPTG}} &= H_0^{(\pi)}\mathcal{M}_{\text{AL},0} \\ &= (1 - \omega_{\text{AL}})H_{0,\text{ac}}^{\text{CPTG}} + \omega_{\text{AL}}H_{0,\text{lum}}^{\text{CPTG}} \\ &= 70.3238983675 \pm 0.43 \text{ km s}^{-1} \text{ Mpc}^{-1}. \end{aligned} \quad (137)$$

The uncertainty is half of the declared luminosity-endpoint uncertainty because the present acoustic endpoint is fixed by the registered CPTG acoustic projection. The rate-space result $\mathcal{M}_{\text{AL}}$ inherits the luminosity coordinate's observational conditioning and is therefore data-conditioned, whereas $\omega_{\text{AL}} = 1/2$ is the fixed symmetric comparison prescription.

For nonlinear observables, midpoint formation is performed after each endpoint observable has been evaluated. For example, if the two endpoint distance coordinates are $D_{\text{ac}}(N_\pi)$ and $D_{\text{lum}}(N_\pi)$, then

$$D_{\text{mid}}(N_\pi) = (1 - \omega_{\text{AL}})D_{\text{ac}}(N_\pi) + \omega_{\text{AL}}D_{\text{lum}}(N_\pi), \quad (138)$$

rather than constructing a third branch rate and then re-integrating it as though it were an independent cosmology.

The simple geometric value

$$\mathcal{G}_L^{\text{candidate}} = \frac{10}{9} \quad (139)$$

provides the endpoint candidate

$$H_{0,\text{lum}}^{\text{candidate}} = H_0^{(\pi)} \sqrt{\frac{10}{9}} = 73.1711530416 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (140)$$

Its proximity to the SH0ES-conditioned endpoint remains a diagnostic observation. It is not required to define Eqs. (125)–(130), and it is not promoted here to a microscopic luminosity projector or redshift exponent.

11.4 Common-Phase Horizon and Last-Scattering Coordinates

The branch clocks are compared only at equal geometric phase. For the Planck last-scattering reference

$$z_\star = 1089.92, \quad a_{\text{dec}} = \frac{1}{1 + z_\star} = 9.1665750009 \times 10^{-4}, \quad (141)$$

the CPTG acoustic comparison distance is

$$\chi_\star = \frac{c}{H_0^{(\pi)}} \int_0^{z_\star} \frac{dz}{E_{\text{ac}}(z)} = 13876.8219 \text{ Mpc}. \quad (142)$$

The dimensional prefactor is the native normalization $H_0^{(\pi)}$; the exact acoustic projection enters through E_{ac} . This convention is important because substituting $H_{0,\text{ac}}^{\text{CPTG}}$ into the prefactor would define a different distance coordinate.

The physical radius of the region that becomes the observed last-scattering sphere is

$$L_\star(a) = a\chi_\star. \quad (143)$$

At decoupling,

$$L_\star(z_\star) = 12.72029287 \text{ Mpc}, \quad R_C(z_\star) = 23.64711498 \text{ Mpc}, \quad (144)$$

and therefore

$$\frac{L_\star(z_\star)}{R_C(z_\star)} = 0.53792156 < 1. \quad (145)$$

This is a radius-to-radius comparison at one geometric phase. A diameter-to-radius ratio is not used as a containment criterion [40].

The equality $L_\star = R_C$ occurs at

$$z_{\text{enter}} = 3231.635355, \quad L_\star = R_C = 4.292727257 \text{ Mpc}, \quad (146)$$

with native thermal temperature

$$T_{\text{enter}} = T_0(1 + z_{\text{enter}}) = 8810.483007 \text{ K}. \quad (147)$$

This entry phase marks geometric containment; it is not defined as the beginning of coherence.

The native and exact acoustic clocks at the same two geometric phases are

$$z_{\text{enter}} : \quad t_{\pi} \simeq 48,802 \text{ yr}, \quad t_{\text{ac}} \simeq 106,889 \text{ yr}, \quad (148)$$

$$z_{\star} : \quad t_{\pi} \simeq 268,835 \text{ yr}, \quad t_{\text{ac}} \simeq 536,006 \text{ yr}. \quad (149)$$

These are branch-specific integrations at common N_{π} , not a cross-branch synchronization condition. Together with Eqs. (125)–(126), they complete the reference architecture of one native branch and two projected branch clocks.

Using the Planck acoustic-angle anchor $100\theta_{\star} = 1.04110$, the distance above corresponds arithmetically to

$$r_{\text{ac}} = \theta_{\star}\chi_{\star} = 144.4716 \text{ Mpc}, \quad (150)$$

which is close to the Planck-inferred $r_{\star} = 144.43 \pm 0.26 \text{ Mpc}$ [2]. This is a comparison-coordinate reconstruction from the adopted angular anchor, not an independent prediction of $\theta_{\star}$.

For a spatially flat comparison branch, the transverse comoving distance is

$$D_M(z) = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}, \quad (151)$$

the radial Hubble distance is

$$D_H(z) = \frac{c}{H_0 E(z)}, \quad (152)$$

and the volume-averaged acoustic distance is

$$D_V(z) = \left[z D_M(z)^2 D_H(z) \right]^{1/3}. \quad (153)$$

BAO likelihoods compare acoustic ratios rather than dimensional distances alone. The primary coordinates are

$$\frac{D_M(z)}{r_d}, \quad \frac{D_H(z)}{r_d}, \quad \frac{D_V(z)}{r_d}. \quad (154)$$

For anisotropic BAO or ShapeFit-style comparisons,

$$q_{\parallel} = \frac{D_H(z)/r_d}{[D_H(z)/r_d]_{\text{fid}}}, \quad (155)$$

$$q_{\perp} = \frac{D_M(z)/r_d}{[D_M(z)/r_d]_{\text{fid}}}, \quad (156)$$

$$q_{\text{iso}} = (q_{\perp}^2 q_{\parallel})^{1/3}, \quad (157)$$

$$q_{\text{ap}} = \frac{q_{\parallel}}{q_{\perp}}. \quad (158)$$

When an absolute BAO-ruler normalization is required by the present compressed implementation, the fixed working comparison normalization is

$$r_d = 148.998491 \text{ Mpc}. \quad (159)$$

In the existing compressed BAO analysis it is a fitted comparison-layer normalization, not a native geometric invariant. The separate quarter-ruler relation is dimensionless and is not used to reclassify this fitted absolute normalization as a native quantity.

Supernova comparisons use

$$D_L(z) = (1+z)D_M(z), \quad \mu(z) = 5 \log_{10} \left[\frac{D_L(z)}{\text{Mpc}} \right] + 25. \quad (160)$$

The supernova comparison is therefore a distance-shape comparison after standardization, intercept treatment, calibration handling, and covariance weighting [5]. BAO comparisons are made through survey-defined distance/ruler ratios and AP coordinates [7, 8, 6]. In both cases CPTG supplies the geometric distance relation first; the observational pipeline determines which comparison coordinates are evaluated.

12 Primordial Network Implementation and Verification Record

The microscopic and primordial comparison layer is tested after its coordinates are fixed. The following validations therefore remain downstream of the source, reaction-observable, network-response, and mass-seven definitions above; they do not define those coordinates.

12.1 Commissioning-Channel Reaction-Observable Verification

The fixed zero-energy commissioning normalization is compared with the Solar Fusion II LUNA-extrapolated value $S_{12}(0) = 0.216 \pm 0.010 \text{ eV b}$, giving a signed residual of -0.149960σ [16, 15]. At $E_0 = 6.64 \text{ keV}$, the fixed finite-energy coordinate is compared with $0.252 \pm 0.018 \text{ eV b}$, or -0.020433σ [16, 15].

The later universal-reaction audit provides a direct empirical check of this observable layer without changing its normalization. The complete Mossa/LUNA replay places all 13 published S -factor points from 32.4 to 262.9 keV inside their combined one-standard-deviation intervals and all 17 tabulated thermonuclear rates from $T_9 = 0.001$ to 10 inside their published asymmetric one-standard-deviation bounds [17, 22]. The same fixed projection is consistent with the Tišma and Turkat covariance-weighted shape tests, while the three-point Dhibar dataset remains strongly discordant because of its highest-energy point; no theory parameter is changed to absorb that discrepancy [18, 19, 20].

Observable test	CPTG result	External comparison	Classification
Zero-energy $S_{12}(0)$	0.2145003969219307 eV b	$0.216 \pm 0.010 \text{ eV b}$	-0.149960σ
Solar-Gamow $S_{12}(6.64 \text{ keV})$	0.2516322073981925 eV b	$0.252 \pm 0.018 \text{ eV b}$	-0.020433σ ; external shape
Mossa/LUNA S -factor grid	13/13 inside combined 1σ	32.4–262.9 keV	max. $ z_S = 0.951713$
Mossa/LUNA rate grid	17/17 inside published interval	$T_9 = 0.001\text{--}10$	max. $ \tilde{z}_R = 0.959051$
91 keV pivot	0.7931440807 eV b	$0.799 \pm 0.018 \text{ eV b}$	-0.3253σ
Covariance shape checks	Tišma $\chi^2/\nu = 0.2325$; Turkat 1.1137	fixed projection	consistency diagnostics
Dhibar three-point check	$\chi^2/\nu = 15.5173$	fixed projection	discordant; retained

Table 12: Commissioning-channel reaction-observable comparisons. The zero-energy normalization is the CPTG-derived coordinate. Finite-energy and thermal rows retain their declared external energy-shape information. The Mossa/LUNA interval counts and covariance quantities are validation metrics, not additional theory parameters.

12.2 PRIMAT-Native Baryon, Radiation, and Mass-Seven Validation

PRIMAT v0.3.2 is the primary numerical authority for the present full-coordinate primordial-network test [14]. The completed, prospectively preregistered large-network protocol first calibrates the native radiation response with two gate-off rows. This abundance-independent calibration is a native translation step used to implement the already-fixed CPTG radiation coordinate; it is not an independent PRIMAT prediction of that target. The reported values

$$N_{\text{eff}}(\Delta N_{\text{eff}} = 0) = 3.0439772985579183, \quad (161)$$

$$N_{\text{eff}}(\Delta N_{\text{eff}} = 0.01) = 3.0539772985579177 \quad (162)$$

give

$$\frac{dN_{\text{eff}}}{d(\Delta N_{\text{eff}})} = 0.9999999999999343. \quad (163)$$

The prospectively frozen native input required to implement the already-defined CPTG radiation coordinate is therefore

$$\Delta N_{\text{eff}}^{\text{CPTG}} = -0.07539322509392317, \quad (164)$$

set before target abundance exposure. At the joint baryon-plus-radiation target

$$\left((\Omega_b h^2)_{\text{BBN}}, N_{\text{eff}}^{\text{CPTG}} \right) = (0.021898765370, 2.968584073464), \quad (165)$$

PRIMAT returns

$$N_{\text{eff}}^{\text{PRIMAT}} = 2.968584073463995, \quad (166)$$

an absolute coordinate error of only 4.88×10^{-15} . This residual is a native translation/mapping-closure metric after the frozen abundance-independent translation; it is not an independent PRIMAT prediction of $N_{\text{eff}}^{\text{CPTG}} = p_{\text{ac}}$. No deuterium, helium, or lithium abundance is used to construct the radiation mapping.

The observational background controls are

$$10^6(\text{D/H})_{\text{obs}} = 25.08 \pm 0.29, \quad Y_{p,\text{obs}} = 0.245 \pm 0.003 \quad (167)$$

[13]. The lithium value is retained only as a downstream diagnostic and is excluded from the scientific acceptance gates.

Quantity	Native PRIMAT result	Frozen reference or control	Residual / role
N_{eff} , target	2.968584073463995	2.968584073464	4.88×10^{-15} absolute; pass
D/H, gate off	$25.0669253584 \times 10^{-6}$	$(25.08 \pm 0.29) \times 10^{-6}$	-0.045085σ ; pass
Y_p , gate off	0.245760246933	0.245 ± 0.003	$+0.253416\sigma$; pass
Live $A = 7$ survival	0.318366498376680	$1/\pi = 0.318309886183791$	0.0177852% relative; pass
D/H gate disturbance	3.91×10^{-9} relative	$< 10^{-6}$	Selectivity pass
Y_p gate disturbance	3.86×10^{-12} relative	$< 10^{-6}$	Selectivity pass
$^3\text{He}/\text{H}$ gate disturbance	5.59×10^{-10} relative	$< 10^{-6}$	Selectivity pass
$^7\text{Li}/\text{H}$, gate off	$5.2735647904 \times 10^{-10}$	$(1.45 \pm 0.25) \times 10^{-10}$	$+15.2943\sigma$; diagnostic only
$^7\text{Li}/\text{H}$, gate on	$1.6789263563 \times 10^{-10}$	$(1.45 \pm 0.25) \times 10^{-10}$	$+0.915705\sigma$; diagnostic only

Table 13: Completed prospectively preregistered PRIMAT native-coordinate implementation and downstream response validation at the CPTG transported baryon and radiation coordinate. D/H and Y_p are preregistered background gates; the mass-seven survival and non-mass-seven disturbance rows test the live transport law; lithium abundance is report-only and does not enter the hard decision [14, 13, 23].

With the gate enabled, the non-mass-seven native outputs are

$$\text{D/H}_{\text{gated}} = 2.5066925456419876 \times 10^{-5}, \quad (168)$$

$$Y_{p,\text{gated}} = 0.24576024693245754, \quad (169)$$

$$(^3\text{He}/\text{H})_{\text{gated}} = 1.0503519364716970 \times 10^{-5}. \quad (170)$$

The network-realized live mass-seven survival differs from the exact isolated coordinate $\mathcal{S}_7 = 1/\pi$ because ordinary native production and destruction continue during the transport window; the measured deviation is therefore a genuine network-response test rather than a built-in algebraic identity.

All 23 preregistered scientific gates pass. The result establishes completed prospectively preregistered PRIMAT-native implementation and mapping closure for the jointly transported CPTG baryon and radiation coordinate, together with downstream native-network response and selectivity validation of the locked live $A = 7$ abundance-transport law for the tested PRIMAT v0.3.2 large-network configuration. It does not imply that PRIMAT independently derives $N_{\text{eff}}^{\text{CPTG}} = p_{\text{ac}}$ or $\tau_7 = \ln \pi$, and it does not convert τ_7 into a stellar depletion factor, a laboratory cross section, a post-processing abundance correction, or a fifth coordinate of the four-sector nuclear simplex.

12.3 PRIMAT-Native First-Order Reaction Pushforward Validation

The network-conditioned coordinate in Equation (59) is tested separately from the mass-seven transport law. PRIMAT’s completed full-reaction authority contains 20,550/20,550 committed native rows: six unperturbed baseline rows plus $428 \times 6 \times 8 = 20,544$ reaction–anchor–branch perturbation rows. Thus

$$6 + 428 \times 6 \times 8 = 20,550. \quad (171)$$

A separate native source/Jacobian campaign constructs 84 first-order variational predictions from fourteen mapped light-sector reactions at the same six anchors. The prediction table is frozen before the authoritative perturbed-endpoint coefficients are opened, so the endpoint response is an evaluation target rather than a fitting input [22, 14].

The endpoint reference at each reaction–anchor pair is independently extracted from the completed

eight-branch ladder through

$$\Delta \ln Y = c_1 x + c_2 x^2, \quad x = -\ln f_r, \quad (172)$$

with zero intercept. For a primary resolved first-order response, $\|\mathbf{c}_{1,\text{ref}}\| \geq 10^{-3}$, the endpoint-vector discrepancy is

$$\epsilon_{\text{vec}} = \frac{\|\mathbf{c}_{1,\text{var}} - \mathbf{c}_{1,\text{ref}}\|}{\|\mathbf{c}_{1,\text{ref}}\|}, \quad (173)$$

and the direction metric is the ordinary cosine

$$\cos \theta = \frac{\mathbf{c}_{1,\text{var}} \cdot \mathbf{c}_{1,\text{ref}}}{\|\mathbf{c}_{1,\text{var}}\| \|\mathbf{c}_{1,\text{ref}}\|}. \quad (174)$$

The frozen gates are $\epsilon_{\text{vec}} \leq 0.02$ and $\cos \theta \geq 0.999$. Independent BDF and Radau propagations are also required to agree under the separately frozen norm-based cross-solver relative-difference metric, $\epsilon_{\text{BDF/Radau}} \leq 5 \times 10^{-4}$, with cross-solver direction cosine at least 0.999999.

Validation quantity	PRIMAT result	Interpretation / gate
Frozen source/Jacobian predictions	84/84	Completed before endpoint-reference exposure
Primary resolved tests	48/48 pass	$\ \mathbf{c}_{1,\text{ref}}\ \geq 10^{-3}$
Primary + secondary resolved cases	66/84	Response norm at least 10^{-4} ; secondary reporting floor
Worst primary endpoint-vector discrepancy	3.6762871×10^{-4} (0.0367629%)	Frozen requirement ≤ 0.02
Minimum primary endpoint direction cosine	0.9999999921	Frozen requirement ≥ 0.999
Maximum BDF/Radau cross-relative difference	5.8248×10^{-7}	Frozen requirement $\leq 5 \times 10^{-4}$
Maximum source-direction residual	3.17×10^{-16}	Frozen requirement $< 10^{-10}$
Maximum baryon-conservation residual	4.44×10^{-16}	Frozen requirement $< 10^{-10}$

Table 14: Completed PRIMAT first-order native source/Jacobian validation of the network-conditioned nuclear reaction-response pushforward. The authoritative endpoint coefficients are not used to construct the frozen variational predictions [22, 14].

For the commissioned $D(p, \gamma)^3\text{He}$ channel, the largest six-anchor endpoint-vector discrepancy across either variational solver is approximately 1.62×10^{-4} (0.0162%), with direction cosine effectively unity. This result is especially important for comparison-coordinate interpretation: it validates the route

$$\mathbf{s}_r \longrightarrow \mathbf{d}_{r,\mathcal{N}}^{(1)} \longrightarrow \mathbf{y}_{r,\mathcal{N}}^{(1)} \longrightarrow \mathcal{C}_{r,\mathcal{N}}^{(1)}, \quad (175)$$

not the shortcut $\mathbf{s}_r \equiv \mathbf{c}_1$.

ParthENoPE 3.0 supplies the completed second-authority transfer in the shared endpoint space. Its archived campaign contains 695/695 native rows, 338/338 matched pairs, and 84/84 complete ladders; the declared Reaction-20 transfer has direction cosine 0.999979437501036, with all component-ratio and six-anchor stability gates satisfied [22, 24]. This supports the same comparison architecture while preserving code-local currents, source kernels, Jacobians, normalizations, and solver trajectories as network-specific objects. PRyMordial full-native authority constitutes a separate validation lane and is not required to define the PRIMAT first-order coordinate reported here.

13 CMB Implementation and Full-Stack Verification Record

The definitions in the preceding CMB construction sections remain authoritative. The material below records how those fixed coordinates were exercised numerically; it does not redefine them.

13.1 Curvature-Transport Spectrum Bridge Verification

The bridge was exercised through CAMB and multiple Planck likelihood coordinates. The NPIPE TTTEEE stress ladder is monotonic from the identity row $f_r = 1$ through $f_r = 0.990$, where the accumulated shift is $\Delta\chi^2 = +2.923593$. NPIPE TT, TE, EE, and TTTEEE split sectors pass monotonically. Plik TT, EE, and TTTEEE also pass monotonically; the TE sector contains a localized diagnostic inversion with its minimum at $f_r = 0.996$ and $\Delta\chi^2 = -0.445906$. In the near-identity full stack including lensing, $f_r = 0.999$ gives $\Delta\chi^2 = +0.350109$. These f_r values are stress coordinates only, and the summed χ^2 values are bridge smoke metrics rather than replacements for the profiled CMB validation reported later in this article [41].

13.2 CMB Map-Envelope Verification

Under the declared Planck SMICA comparison, the fitted CPTG map has residual RMS $18.218928 \mu\text{K}$ and valid-sky correlation 0.983434 . Across the five principal Planck component maps, the mean CPTG-minus-Planck full-band residual-RMS gap is only $0.023518 \mu\text{K}$. The SMICA full/half-mission/odd-even split test gives a mean gap of $0.023891 \mu\text{K}$ with a split-to-split span of $0.000690 \mu\text{K}$. The null-envelope audit is the uniqueness control: mean residual RMS is $18.230 \mu\text{K}$ for the Planck benchmark and $18.253 \mu\text{K}$ for the locked CPTG envelope, while shuffled-spectrum nulls rise to $95.461 \mu\text{K}$. Thus phase locking alone is insufficient to generate the observed near-degeneracy; the angular-power envelope must retain the relevant acoustic structure [42].

The map-envelope verification retains the same narrow scope stated in the construction section: it tests a data-conditioned real-map projection with the observed phase scaffold declared explicitly, not a native prediction of the exact random CMB phase realization.

13.3 Fixed-Row Likelihood Verification Protocol

The cleaned CMB branch is evaluated with fixed CPTG cosmology and fixed comparison gates. The comparison statistic is

$$\Delta\chi^2 = \chi_{\text{CPTG,locked}}^2 - \chi_{\text{control}}^2, \quad (176)$$

where the control row is evaluated under the same likelihood, nuisance treatment, profiling or sampling protocol, and data combination.

For this fixed-row robustness audit, the comparison categories are

$$\Delta\chi^2 \leq 5 \implies \text{control level}, \quad (177)$$

$$5 < \Delta\chi^2 \leq 10 \implies \text{watch/stress}, \quad (178)$$

and

$$\Delta\chi^2 > 10 \implies \text{decisive stress}. \quad (179)$$

These are audit thresholds, not universal frequentist confidence levels. Their scale is empirically motivated by established CMB likelihood-robustness studies. In the Planck 2018 high- ℓ validation, evaluating TE

relative to a TE-only best-fit cosmology changes the fit by about $|\Delta\chi^2| = 4.6$, while alternative polarization-efficiency treatments produce changes of order 9–10; a much more flexible 18-parameter leakage correction changes χ^2 by 17.5 but is not adopted because the improvement is comparable to the number of added degrees of freedom and exceeds the expected instrumental-uncertainty structure [3]. These published robustness scales motivate treating few-unit changes as control-level variation, intermediate 5–10 changes as a watch/stress region, and double-digit degradation as a qualitatively stronger stress signal. Recent Plik/CamSpec-NPIPE comparisons likewise find very similar cosmological solutions across likelihood choices, with remaining differences reduced further when compatible ground-based CMB information is added [37]. The 5 and 10 boundaries therefore define pragmatic fixed-row robustness categories, not parameter-estimation confidence contours or universal model-selection thresholds.

All CMB comparisons in this section keep the locked gates fixed:

$$\mathcal{D}_c = 0.9735353159758136, \quad (180)$$

$$\mathcal{A}_s = \frac{\pi}{3}, \quad (181)$$

$$\mathcal{T}_\tau = \mathcal{G}_T^2, \quad (182)$$

and

$$A_{\text{lens}} = 1. \quad (183)$$

No result in this section retunes the cold-sector transport, scalar-amplitude gate, reionization-optical-depth gate, or lensing-amplitude setting.

13.4 Validation Milestone Summary

The CMB comparison layer is tested through high- ℓ likelihoods, posterior-proxy stability sampling, low- ℓ polarization, lensing diagnostic channels, and combined likelihood stacks. Table 15 summarizes the validation status of the locked cleaned $\mathcal{G}_T^2$ comparison row.

The low- ℓ , lensing, and MCMC rows are reported as diagnostics where appropriate. The principal validation claim rests on the fixed-row likelihood/profile behavior and exact gate closure; MCMC likelihood stability is supporting evidence and is not relabeled as conventional chain convergence when the reported $R - 1$ values exceed the standard Cobaya stopping criterion.

Validation layer	Result summary	Status
High- ℓ profile robustness	Plik TTTEEE, SPT-3G TEEE, NPIPE CamSpec TTTEEE: mean median shift +2.221794; maximum median shift +4.501564; CamSpec worst repeat +5.013677.	Validation pass; CamSpec watch
Posterior-proxy stability check	Plik TTTEEE and SPT-3G TEEE: mean shift +1.6722; maximum shift +1.9233.	Control-level stability; convergence not claimed
Low- ℓ and lensing diagnostic stability	Planck low- ℓ EE, Planck lensing, ACT DR6 lensing: Planck mean +1.589; maximum Planck shift +2.149.	Diagnostic pass
Full-stack profile validation	Plik+lowE+lensing, NPIPE CamSpec+lowE+lensing, SPT+ACT: mean primary shift +2.101464; maximum primary shift +3.669953; CamSpec worst repeat +5.990520, in the watch/stress band.	Validation pass; CamSpec watch repeat retained
Combined-stack posterior stability	Plik+lowE+lensing, full NPIPE CamSpec+lowE+lensing, SPT+ACT: mean median shift +0.096642; maximum median shift +0.737534; maximum reported $R - 1 = 0.143463$.	Control-level likelihood stability; convergence not claimed
Final validation consolidation	Consolidated profile, posterior-stability, and gate status: maximum designated consolidation statistic $ \Delta\chi^2 = 4.824743$ with $A_{\text{lens}} = 1$; the separate CamSpec worst-repeat diagnostic is 5.990520.	Profile validation with supporting stability diagnostics; worst repeat in watch band
Amplitude/reionization-depth gate closure	Gates $\mathcal{A}_s = \pi/3$ and $\mathcal{T}_\tau = \mathcal{G}_T^2$: exact closure with maximum relative difference 0.	Numerical validation pass

Table 15: CMB validation milestones for the locked cleaned $\mathcal{G}_T^2$ comparison row.

13.5 High- ℓ Profile Robustness

The locked cleaned row is tested against distinct high- ℓ likelihood implementations using repeated nuisance-profile comparisons. The aggregate designated median shift is

$$\langle \Delta\chi_{\text{median}}^2 \rangle = 2.221794, \quad (184)$$

and the maximum designated high- ℓ median shift is

$$\max(\Delta\chi_{\text{median}}^2) = 4.501564. \quad (185)$$

The Plik and SPT-3G channels are stable at control level. NPIPE CamSpec remains control-level by the comparison statistic while showing profile sensitivity associated with nuisance and calibration basin structure in the full CamSpec implementation.

13.6 Plik and SPT Posterior-Proxy Stability

The designated Plik and SPT-3G high- ℓ channels are also tested with MCMC posterior-proxy sampling. All four runs terminate before the configured sample cap, but the final $R - 1$ values reported below exceed Cobaya’s default $R - 1 = 0.01$ stopping criterion. The runs therefore establish posterior-proxy likelihood stability under the configured campaign settings; conventional chain convergence is not claimed here [30, 31].

Target	CPTG proxy χ^2	Control proxy χ^2	Shift	CPTG samples	Status
Plik TTTEEE	2344.4088	2342.4855	+1.9233	33520	Control-level
SPT-3G TEEE	1112.9661	1111.5450	+1.4211	18680	Control-level

Table 16: Plik/SPT posterior-proxy likelihood-stability comparison for the locked cleaned row.

The corresponding control chains contain 57600 accepted Plik samples and 28120 accepted SPT samples. The final mean $R - 1$ values are

$$R_{\text{Plik,CPTG}} - 1 = 0.029169, \quad R_{\text{Plik,control}} - 1 = 0.029464, \quad (186)$$

$$R_{\text{SPT,CPTG}} - 1 = 0.028239, \quad R_{\text{SPT,control}} - 1 = 0.025843. \quad (187)$$

Cobaya’s default stopping criterion is $R - 1 < 0.01$, and its documentation emphasizes that this diagnostic is necessary but not sufficient for convergence [31]. The values above therefore should not be read as conventionally converged posterior chains.

The mean and maximum designated likelihood shifts are

$$\langle \Delta \chi^2_{\text{posterior}} \rangle = 1.6722, \quad \max(\Delta \chi^2_{\text{posterior}}) = 1.9233. \quad (188)$$

The Plik and SPT posterior-proxy likelihood shifts therefore remain inside the control-level region. This is a posterior-stability diagnostic rather than a conventional convergence claim.

13.7 Low- ℓ and Lensing Diagnostic Stability

The cleaned row is tested in low- ℓ polarization and lensing channels without retuning the reionization optical depth and without increasing A_{lens} . Table 17 reports the diagnostic stability result.

Target	CPTG median χ^2	Control median χ^2	Median shift	Status
Planck low- ℓ EE	396.550	395.521	+1.029	Control-level
Planck 2018 lensing	10.936	8.88949	+2.149	Control-level
ACT DR6 lensing	14.7052	13.9677	+0.7375	Control-level

Table 17: Low- ℓ and lensing diagnostic stability for the locked cleaned row.

For the designated Planck low- ℓ and Planck lensing rows, the mean median shift is

$$\langle \Delta \chi^2_{\text{lowE/lensing}} \rangle = 1.589, \quad (189)$$

and the maximum designated median shift is

$$\max(\Delta \chi^2_{\text{lowE/lensing}}) = 2.149. \quad (190)$$

Thus the cleaned CMB reionization-depth relation

$$\tau_{\text{reio,eff}} = 0.05893087593426084 \quad (191)$$

remains compatible with the low- ℓ and lensing diagnostic channels under the same locked comparison layer.

13.8 Combined-Stack Profile Validation

The strongest completed CMB validation layer is the locked combined-stack profile audit. This test combines high- ℓ , low- ℓ , and lensing likelihoods where available and keeps the same locked CMB row. The

audit treats NPIPE CamSpec as a distinct Planck PR4 likelihood implementation applied to overlapping Planck sky information; implementation independence is not statistical data independence. When nuisance-marginalized CamSpec-NPIPE-lite is not available in the likelihood environment, full NPIPE CamSpec is evaluated with a best-of- N profile-sensitive protocol [37].

All enabled combined-stack profile runs succeed:

$$44/44 = 1. \quad (192)$$

There are no timeouts and no run failures. The designated combined-stack results are given in Table 18.

Stack	Protocol	Primary shift	Median shift	Worst shift
Plik TTTEEE + lowE + Planck lensing	Median profile	+1.896906	+1.896906	+2.418011
NPIPE CamSpec + lowE + Planck lensing	Best-of- N profile	+3.669953	+4.824743	+5.990520
SPT-3G TEEE + ACT DR6 lensing	Median profile	+0.737534	+0.737534	+0.737534

Table 18: Locked combined-stack CMB profile validation. NPIPE CamSpec is treated as a distinct PR4 likelihood implementation using full CamSpec best-of- N when CamSpec-NPIPE-lite is unavailable; it is not statistically independent of other Planck sky analyses.

The primary-shift mean is

$$\langle \Delta\chi^2_{\text{primary}} \rangle = 2.101464, \quad (193)$$

and the largest primary shift is

$$\max(\Delta\chi^2_{\text{primary}}) = 3.669953. \quad (194)$$

This places Plik, full NPIPE CamSpec, and the ground high- ℓ +lensing stack inside the control-level region under the same fixed CMB row. The CamSpec worst repeat, 5.990520, lies in the explicit watch/stress band, while the primary best-of- N CamSpec comparison remains control-level. This is interpreted as successful combined-stack profile validation with CamSpec retained as a distinct but profile-sensitive PR4 likelihood implementation.

13.9 Combined-Stack Posterior Stability Diagnostic

The same locked CMB comparison row is also carried through posterior sampling across the Plik, full NPIPE CamSpec, and SPT/ACT stack combinations. The diagnostic keeps $A_{\text{lens}} = 1$ and does not retune τ_{reio} , $\mathcal{D}_c$, $\mathcal{A}_s$, or $\mathcal{T}_\tau$. All 6/6 enabled MCMC jobs complete without timeouts.

Stack	Median shift	Max final $R - 1$	Classification
Plik TTTEEE + lowE + Planck lensing	-0.2238004	0.121641	Likelihood stable; convergence not claimed
NPIPE CamSpec + lowE + Planck lensing	-0.2238066	0.143463	Likelihood stable; convergence not claimed
SPT-3G TEEE + ACT DR6 lensing	+0.7375340	0.108998	Likelihood stable; convergence not claimed

Table 19: Locked combined-stack CMB posterior stability diagnostic. Full NPIPE CamSpec is used when CamSpec-lite is unavailable.

The median-shift mean and maximum are

$$\langle \Delta\chi^2_{\text{combined posterior}} \rangle = 0.0966423333, \quad \max(\Delta\chi^2_{\text{combined posterior}}) = 0.737534. \quad (195)$$

The maximum reported final diagnostic is

$$\max(R - 1) = 0.143463. \quad (196)$$

The likelihood shifts are therefore control-level while preserving the fixed acoustic-transport gates. Because the reported $R - 1$ values are 0.109–0.143, this layer is retained as a posterior stability diagnostic and not as a conventional convergence claim [31].

13.10 Numerical Amplitude and Reionization-Depth Gate Closure

The locked comparison gates are also checked as an explicit closure calculation against the final CMB row. This audit does not rerun any likelihood, does not change the CMB row, and does not alter the validation protocol. It verifies the arithmetic closure of the native-to-effective map,

$$\Omega_{c,\text{eff}} h^2 = \Omega_{c,\text{CPTG}} h^2 \mathcal{D}_c, \quad (197)$$

$$A_{s,\text{eff}} = A_{s,\text{CPTG}} \frac{\pi}{3}, \quad (198)$$

$$\tau_{\text{reio},\text{eff}} = \tau_{\text{reio},\text{CPTG}} \mathcal{G}_T^2. \quad (199)$$

Using the frozen pre-map comparison inputs,

$$\Omega_{c,\text{CPTG}} h^2 = 0.120885025627, \quad (200)$$

$$A_{s,\text{CPTG}} = 2.04 \times 10^{-9}, \quad (201)$$

$$\tau_{\text{reio},\text{CPTG}} = 0.066, \quad (202)$$

the closure audit gives

$$0.120885025627 \mathcal{D}_c = 0.1176858416205258, \quad (203)$$

$$2.04 \times 10^{-9} \left(\frac{\pi}{3} \right) = 2.136283004441059 \times 10^{-9}, \quad (204)$$

$$0.066 \left(\mathcal{G}_T^2 \right) = 0.05893087593426084. \quad (205)$$

These are exactly the locked effective values used in the CMB comparison row. The maximum relative closure difference is

$$0. \quad (206)$$

The result closes the numerical bookkeeping of the scalar-amplitude and reionization-optical-depth gates. The preceding geometric section states these gates as CPTG curvature-response projections: $\mathcal{A}_s = \pi/3$ is the scalar projection of the π -curvature branch, and $\mathcal{T}_\tau = \mathcal{G}_T^2$ is the two-step visibility contraction of the acoustic transport projection.

13.11 Coupled Gate Perturbation

A coupled p_{ac} -gate perturbation tests whether $\mathcal{G}_T^2$ behaves as a coherent transport relation rather than as an isolated numerical value. The perturbation varies p_{ac} around the locked value, holds n_s and N_{eff} fixed to isolate the gate-coupling layer, and applies

$$\mathcal{G}_T(p_{\text{ac}}) = \frac{p_{\text{ac}}}{\pi}, \quad (207)$$

$$\mathcal{D}_c(p_{\text{ac}}) = \sqrt{\mathcal{G}_T(p_{\text{ac}})} \left[1 + \frac{1}{10\pi} \left(\frac{\pi}{3} - 1 \right) \right], \quad (208)$$

$$\tau_{\text{reio,eff}}(p_{\text{ac}}) = \tau_{\text{reio,CPTG}} \mathcal{G}_T(p_{\text{ac}})^2. \quad (209)$$

The result is summarized in Table 20.

Rank	p_{ac} shift	Mean target-median shift	Max target-median shift	Status
1	+0.005	+2.212364	+4.574011	Control-level
2	0.000, locked	+2.297587	+4.677919	In basin
3	+0.010	+2.299781	+4.901645	In basin
4	−0.005	+2.584480	+5.078886	Mild stress
5	−0.010	+3.193148	+5.921761	Stress
6	+0.020	+3.543665	+7.271139	Stress
7	−0.020	+5.479216	+9.497230	Poor

Table 20: Coupled p_{ac} -gate perturbation. The locked row remains inside the coupled-gate basin.

The locked row ranks second and is only

$$0.085223 \quad (210)$$

above the best aggregate row in the perturbation audit. This supports $\mathcal{G}_T^2$ as a coherent coupled-gate relation inside the locked CMB acoustic-transport layer.

13.12 CMB Implementation Boundary

The CMB implementation status is

CPTG passes the locked cleaned CMB comparison layer

through high- ℓ profile robustness and

combined-stack profile validation, with supporting (211)

posterior stability and low- ℓ /lensing diagnostics,

plus exact gate closure, with $A_{\text{lens}} = 1$.

The completed CMB evidence supports the fixed row without reionization-optical-depth retuning or phenomenological lensing enhancement. The posterior jobs show control-level likelihood shifts but do not meet Cobaya’s default $R - 1 < 0.01$ convergence criterion, so conventional posterior convergence is not

assigned to this implementation record. The gate-closure audit independently verifies exact numerical closure of $\Omega_{c,\text{eff}}h^2$, $A_{s,\text{eff}}$, and $\tau_{\text{reio},\text{eff}}$ against the locked row.

14 DESI DR1 Adapter and Diagnostic Record

The DESI DR1 coordinate comparison is organized into three distinct layers. The DESI DR1 BAO measurements and full-shape analyses provide the published context for these coordinate-level diagnostics [7, 8, 9, 10]. The first layer is the compressed ShapeFit coordinate comparison using the official DESI DR1 HDF5 likelihood containers associated with the full-shape and BAO clustering products [11, 12]. The second layer is the BAO quarter-ruler coordinate-wrapper diagnostic. The third layer is a full-shape spectrum-shell AP/growth diagnostic using **desilike** coordinate behavior and the native HDF5 spectrum vectors.

The third layer is kept separate from the raw full-shape validation claim. It is a completed exploratory spectrum-shell diagnostic using the declared native HDF5 spectrum vectors and **desilike** coordinate behavior; a raw CPTG full-shape power-spectrum likelihood validation is not claimed here.

14.1 Compressed ShapeFit Coordinate Comparison

The ShapeFit HDF5 containers are checked with two reference vectors. The identity vector, where the model vector equals the observed compressed vector, gives

$$\chi_{\text{identity}}^2 = 0. \quad (212)$$

The no-shift reference vector is

$$(q_{\text{iso}}, q_{\text{ap}}, d_m, d_f) = (1, 1, 0, 1), \quad (213)$$

which gives

$$\chi_{\text{no shift}}^2 = 38.70438441. \quad (214)$$

The locked CPTG compressed-coordinate map follows the **desilike**/ShapeFit AP convention [12] and converts the transverse and radial coordinates into the ShapeFit basis by

$$q_{\text{iso}} = (q_{\perp}^2 q_{\parallel})^{1/3}, \quad (215)$$

and

$$q_{\text{ap}} = \frac{q_{\parallel}}{q_{\perp}}. \quad (216)$$

The ShapeFit growth coordinate is used as a ratio to the fiducial growth rate,

$$d_f = \frac{f_{\text{CPTG}}}{f_{\text{fid}}}. \quad (217)$$

The tilt coordinate is held fixed at

$$d_m = 0. \quad (218)$$

This is an adapter choice for the present compressed test, not a derived CPTG prediction. A nonzero d_m should be introduced only after an explicit CPTG spectral-shape mapping is derived.

Thus the locked compressed ShapeFit vector is

$$\mathbf{x}_{\text{CPTG}} = \left((q_{\perp}^2 q_{\parallel})^{1/3}, \frac{q_{\parallel}}{q_{\perp}}, 0, \frac{f_{\text{CPTG}}}{f_{\text{fid}}} \right). \quad (219)$$

For the DESI-like fiducial comparison, the direct HDF5 adapter gives

$$\chi_{\text{CPTG}}^2 = 35.19040980558847, \quad (220)$$

with

$$\chi_{\nu}^2 = 1.466267075232853, \quad (221)$$

and

$$\Delta\chi^2 = -3.51397460441153 \quad (222)$$

relative to the no-shift reference. For the Planck-like fiducial cross-check, the corresponding result is

$$\chi_{\text{CPTG}}^2 = 38.46800119, \quad (223)$$

$$\chi_{\nu}^2 = 1.60283338, \quad (224)$$

and

$$\Delta\chi^2 = -0.23638322. \quad (225)$$

This is a compressed coordinate-level pass. It is not a raw DESI power-spectrum likelihood validation.

14.2 BAO Quarter-Ruler Coordinate Wrapper

The BAO diagnostic uses the quarter-ruler transport relation

$$\mathcal{G}_T = \frac{p_{\text{ac}}}{p_C}. \quad (226)$$

The corresponding ruler ratio is

$$\frac{r_{d,\text{comp}}}{r_{d,\text{fid}}} = \mathcal{G}_T^{-1/4} = 1.014261942163, \quad (227)$$

with the no-distance-shift AP compression scale

$$Q_{\downarrow} = \mathcal{G}_T^{1/4} = 0.985938600700. \quad (228)$$

At unity, the BAO compression wrapper gives

$$\chi_{\text{BAO}}^2(q=1) = 20.12426210852749. \quad (229)$$

The official BAO wrapper exposes Ω_m , $\log \mathcal{L}$, and $\log \pi$ as runtime or pipeline parameters, but it does not expose q_{iso} , $q_{\parallel}$, or $q_{\perp}$ as direct runtime inputs. Therefore the non-unity AP/ruler response is treated as a coordinate-wrapper diagnostic unless an upstream BAO theory calculator supplies the non-unity coordinates through the official likelihood path.

With the CPTG acoustic value of Ω_m and a free ruler ratio, the best non-boundary coordinate-wrapper result occurs near

$$\frac{r_{d,\text{comp}}}{r_{d,\text{fid}}} \simeq 1.014, \quad (230)$$

with

$$\chi^2 = 13.501774995995627. \quad (231)$$

The fixed quarter-ruler candidate gives

$$\chi_{\text{BAO}}^2(\mathcal{G}_T^{-1/4}) = 13.505973446003. \quad (232)$$

An exponent scan using

$$r_d \mapsto \mathcal{G}_T^\alpha r_d \quad (233)$$

finds

$$\alpha_{\text{best}} = -0.2435, \quad (234)$$

with

$$\chi_{\text{best}}^2 = 13.501376842627. \quad (235)$$

The CPTG quarter candidate is

$$\alpha_{\text{CPTG}} = -\frac{1}{4}, \quad (236)$$

and is only

$$\Delta\chi^2 = 0.0045966 \quad (237)$$

above the grid best. The quarter-ruler relation is therefore a strong coordinate-wrapper diagnostic and remains an open independent derivation target. The near-quarter optimum is reported as a numerical correspondence rather than as a claim that DESI uniquely selected the exponent. It is not stated as a raw official non-unity q runtime likelihood response.

14.3 Full-Shape Spectrum AP/Growth Diagnostic

The full-shape spectrum-shell diagnostic evaluates whether the compressed-coordinate success extends into AP/growth spectrum coordinates. This diagnostic uses the native HDF5 spectrum vectors and `desilike` coordinate behavior. It does not inject a CPTG raw matter power spectrum, does not refit DESI nuisance parameters, and does not constitute a DESI full-shape validation.

The `desilike` AP definitions are [12]

$$q_{\parallel} = \frac{(D_H/r_d)}{(D_H/r_d)_{\text{fid}}}, \quad (238)$$

Layer	Quantity	Result	Classification
ShapeFit identity	χ^2	0	Native null check
ShapeFit no-shift	χ^2	38.70438441	Pipeline baseline
ShapeFit CPTG, DESI-like fiducial	χ^2	35.19040980558847	Coordinate-level pass
ShapeFit CPTG, Planck-like fiducial	χ^2	38.46800119	Cross-check
BAO unity wrapper	χ^2	20.12426210852749	Wrapper baseline
BAO ratio-scan optimum	χ^2	13.501774995995627	Free-ruler-ratio diagnostic
BAO exponent-scan optimum	χ^2	13.501376842627 at $\alpha = -0.2435$	Exponent-scan diagnostic
BAO quarter-ruler candidate	χ^2	13.505973446003	Supported diagnostic; derivation open

Table 21: DESI compressed-coordinate and BAO coordinate-wrapper status. The ratio-scan and exponent-scan optima are distinct pipeline outputs. The quarter-ruler row is a coordinate-wrapper diagnostic, not a raw official non-unity runtime response.

and

$$q_{\perp} = \frac{(D_M/r_d)}{(D_M/r_d)_{\text{fid}}}. \quad (239)$$

The associated AP and isotropic combinations are

$$q_{\text{ap}} = \frac{q_{\parallel}}{q_{\perp}}, \quad (240)$$

and

$$q_{\text{iso}} = q_{\parallel}^{\eta} q_{\perp}^{1-\eta}. \quad (241)$$

In isotropic mode,

$$q_{\parallel} = q_{\perp} = q_{\text{iso}}. \quad (242)$$

The `BAOPowerSpectrumTemplate` default AP mode is the $q_{\parallel}, q_{\perp}$ convention. The HDF5 BAO products are mixed by tracer: BGS and QSO use q_{iso} -type products, while ELG and LRG use $q_{\parallel}, q_{\perp}$ -type products. The ShapeFit d_f coordinate scales the growth-rate ratio rather than inserting an absolute $f\sigma_8$.

The spectrum-shell diagnostics show that the locked quarter-down AP plus CPTG growth ratio improves the global spectrum-shell comparison, but the response is tracer- and multipole-dependent. The full sample improves by

$$\Delta(-2 \log \mathcal{L}) = -184.04148385539065 \quad (243)$$

for the locked global quarter-down-plus-growth case. When split/tracer convention models are allowed, the best spectrum-shell diagnostic model is the BGS/ELG axis-repair plus LRG/QSO locked model, with

$$\Delta(-2 \log \mathcal{L}) = -248.077305. \quad (244)$$

This means the spectrum-shell AP/growth layer is shaped by tracer response, multipole response, AP convention, and spectrum-window handling. It is not a clean universal full-shape validation layer.

Diagnostic channel	Observed response	Interpretation
BAO coordinate wrapper	Quarter ruler improves the compressed BAO coordinate comparison	Strong coordinate-wrapper ruler diagnostic
Full sample spectrum shell	Locked quarter-down AP plus CPTG d_f improves the global shell	Positive but not universal; split models do better
BGS monopole $\ell = 0$	q_{iso} supports quarter-down in low-, mid-, high-, and all- k bins	Ruler channel remains supported
BGS quadrupole $\ell = 2$	q_{iso} wants inverse/up in low-, mid-, high-, and all- k bins	Quadrupole AP/projection/window sensitivity
ELG transverse channel	$q_{\perp}$ wants up broadly, especially through $\ell = 2$ and combined blocks	Broad transverse tracer-response sensitivity
LRG/QSO	Locked quarter-down direction remains supported in low-, mid-, and all- k bins	Stable monopole/ruler-supported sector
Gross window/nuisance structure	BGS/QSO share q_{iso} -type structure, while ELG/LRG share $q_{\parallel}, q_{\perp}$ -type structure	Gross HDF5 structure alone does not explain the split

Table 22: DESI full-shape spectrum-shell diagnostics. These are comparison-layer diagnostics only, not full-shape validation claims.

The local k -bin and multipole diagnostics resolve the spectrum-shell response into tracer-, multipole-, and scale-dependent structure. The BGS monopole supports the CPTG quarter-ruler direction, while the BGS quadrupole reverses direction in every tested k -range, establishing that the reversal is not confined to high k . The ELG transverse $q_{\perp}$ response is likewise broad rather than isolated to high k , while LRG/QSO retain quarter-down support across the tested low-, mid-, and all- k sectors. The full-shape spectrum AP/growth result is retained as a completed exploratory spectrum-shell diagnostic with resolved tracer/multipole structure. This does not weaken the compressed ShapeFit coordinate pass or the BAO/monopole quarter-ruler support.

The DESI status is:

compressed ShapeFit coordinate vector: passed,

BAO quarter-ruler coordinate wrapper: supported, (245)

full-shape spectrum AP/growth: completed exploratory spectrum-shell diagnostic.

15 Late-Time Growth, Lensing, Distance, and Clock Verification Record

The fixed branch can be compared against compressed observational sectors using explicit coordinate conversions. External papers define observational anchors and data products; the χ^2 , mask counts, residuals, and compressed pulls reported below are CPTG pipeline outputs rather than numbers supplied by those papers.

15.1 Compressed Weak-Lensing and CMB-Lensing S_8 Comparison

The late galaxy-lensing coordinate and the CMB-lensing coordinate are different compressed observables:

$$S_{8,\text{late}}^{\text{CPTG}} = 0.788071335232, \quad S_{8,\text{CMBL}}^{\text{CPTG}} = 0.821191065949. \quad (246)$$

For published asymmetric galaxy-lensing uncertainties, this audit uses one rule throughout,

$$\sigma_{\text{sym}} = \frac{\sigma_+ + \sigma_-}{2}. \quad (247)$$

Anchor	Published S_8	σ_{sym}	CPTG residual
DES Y6 3×2pt	0.789 ± 0.012	0.0120	-0.077389σ
HSC-Y3 cosmic shear	$0.776^{+0.032}_{-0.033}$	0.0325	$+0.371426\sigma$
KiDS-1000 MetaCalibration	$0.789^{+0.020}_{-0.024}$	0.0220	-0.042212σ
KiDS-1000 lensfit reference	$0.776^{+0.029+0.002}_{-0.027-0.003}$	0.02812	$+0.429317\sigma$

Table 23: Compressed galaxy weak-lensing comparison rows used by the CPTG S_8 audit [26, 29, 27, 28]. For the lensfit row, statistical and quoted shear-calibration systematic uncertainties are combined in quadrature on each side before applying the common symmetric-error convention.

All four reported late-lensing comparison rows lie within 0.43σ of the transported CPTG coordinate. A purely descriptive inverse-variance summary of these four published analysis rows gives

$$S_{8,\text{WL}}^{\text{desc}} = 0.786438, \quad \sigma_{\text{desc}} = 0.009440, \quad \frac{S_{8,\text{late}}^{\text{CPTG}} - S_{8,\text{WL}}^{\text{desc}}}{\sigma_{\text{desc}}} \simeq 0.173025. \quad (248)$$

This is *not* a joint-survey likelihood or an independent combined measurement. The two KiDS rows are alternative analyses of the same underlying survey data, and the different surveys can have nontrivial covariance and modeling correlations. No combined statistical significance is assigned to this descriptive summary.

For CMB lensing, the appropriate published coordinate is

$$S_8^{\text{CMBL}} \equiv \sigma_8 \left(\frac{\Omega_m}{0.3} \right)^{0.25}, \quad (249)$$

which matches the coordinate class used by the CPTG compressed CMB-lensing map. ACT DR6 reports 0.818 ± 0.022 from ACT lensing alone and 0.813 ± 0.018 from ACT DR6 combined with Planck NPIPE lensing [33].

CMB-lensing reference	Published S_8^{CMBL}	CPTG residual
ACT DR6 lensing only	0.818 ± 0.022	$+0.14505\sigma$
ACT DR6 + Planck NPIPE lensing	0.813 ± 0.018	$+0.45506\sigma$

Table 24: CMB-lensing compressed-coordinate comparison for $S_{8,\text{CMBL}}^{\text{CPTG}} = 0.821191065949$ [33].

The ACT+Planck row contains overlapping lensing information with the ACT-only row, so the two are not inverse-variance combined. The CMB-lensing result is a compressed-coordinate diagnostic, not a full lensing likelihood validation.

15.2 Pantheon+ Supernova Distance-Shape Test

The Pantheon+ comparison uses the published statistical-plus-systematic covariance structure and analytically marginalizes one intercept. The explicit CPTG analysis mask used here retains 1624 non-calibrator rows; that row count is a pipeline output of the stated mask and is not presented as the published Pantheon+ sample size. The primary comparison therefore tests the shape of the supernova distance-redshift relation rather than an absolute H_0 calibration [5].

Model	χ^2	Reduced χ^2	$\Delta\chi^2$
Planck-like Λ CDM	1458.014422	0.898345	0
CPTG raw π branch	1458.058573	0.898373	+0.044151
CPTG acoustic projection	1458.644632	0.898734	+0.630210
DESI-like Λ CDM	1460.490455	0.899871	+2.476033

Table 25: Pantheon+ full-covariance distance-shape comparison. The CPTG mask retains 1624 non-calibrator rows and 1623 effective degrees of freedom after intercept marginalization.

The supernova sector therefore gives a distance-shape pass for the locked CPTG branch. It is not an H_0 calibration claim. The exact mask definition and covariance handling are reproducibility inputs and should accompany the public analysis artifacts.

15.3 Age and Clock Consistency

The native and acoustic ages are the self-synchronized time coordinates of the native branch and its exact acoustic projection. They share the native phase $N_\pi = \ln a$, but each is integrated from its own rate history and the two times are not interchangeable.

For the two standard comparison rows, the Hubble normalization is held fixed at the transported CPTG CMB value,

$$H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (250)$$

and only the matter-fraction comparison coordinate is changed. For a flat Λ CDM comparison background,

$$t_0^{\text{flat}}(\Omega_m; H_0) = \frac{2}{3H_0\sqrt{1-\Omega_m}} \sinh^{-1} \sqrt{\frac{1-\Omega_m}{\Omega_m}}. \quad (251)$$

Using $\Omega_m = 0.315$ for the Planck-like comparison coordinate gives $t_0 = 13.780328$ Gyr, while $\Omega_m = 0.300$ for the DESI-like comparison coordinate gives $t_0 = 13.970356$ Gyr. These rows are therefore internally generated comparison-coordinate ages under a common Hubble normalization, not quotations of the collaborations’ full posterior age inferences.

Coordinate	t_0 Gyr	Difference	Ratio
CPTG raw π geometric	12.754121	−1.026208	0.925531
CPTG acoustic projection E_{ac}	13.885173	+0.104845	1.007608
CPTG CMB density-weighted acoustic diagnostic $E_{\text{ac},\rho}$	13.926455176722689	+0.146127	1.010604
CPTG SH0ES-conditioned luminosity comparison	12.843057	−0.937271	0.931985
Planck-like coordinate ($\Omega_m = 0.315$)	13.780328	0	1
DESI-like coordinate ($\Omega_m = 0.300$)	13.970356	+0.190028	1.013790

Table 26: Present-age comparison across the CPTG native, exact acoustic, SH0ES-conditioned luminosity, and acoustic-density diagnostic time coordinates, together with flat- Λ CDM comparison coordinates. The Planck-like and DESI-like rows use the common acoustic Hubble normalization stated above; the luminosity row uses $H_{0,\text{lum}}^{\text{CPTG}} = 73.17 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

The CPTG acoustic projection lies within about one percent of the Planck-like comparison age. The CMB density-weighted value is the diagnostic defined in Section 11.1; it is not a separate acoustic branch or a second acoustic clock. The native, acoustic, and luminosity times remain branch-specific integrations. The

luminosity value shown here is the SH0ES-conditioned comparison realization of Eq. (127), not a universal clock or an independently initialized cosmology.

16 Implementation Status and Scope Boundaries

For reference use, the CMB implementation status is a fixed-row profile-likelihood recovery record: the evaluated Plik, full NPIPE CamSpec, and SPT/ACT profile stacks remain at control level with $A_{\text{lens}} = 1$ and the acoustic-transport gates fixed. The low- ℓ , lensing, and MCMC layers are supporting diagnostics; conventional posterior convergence is not assigned where the reported $R - 1$ values exceed Cobaya’s default 0.01 stopping criterion [31]. The corresponding operational boundary is explicit: a comparable distinct likelihood/data stack that produced decisive stress under the same fixed row and gates, or recovery that required phenomenological $A_{\text{lens}} > 1$, would invalidate that implementation status.

The fixed-row robustness categories are

$$\Delta\chi^2 \leq 5 \quad (\text{control}), \quad 5 < \Delta\chi^2 \leq 10 \quad (\text{watch/stress}), \quad \Delta\chi^2 > 10 \quad (\text{decisive stress}), \quad (252)$$

and are pragmatic likelihood-robustness categories rather than universal confidence limits [3, 37].

The primordial-network layer fails if the prospectively frozen baryon or radiation coordinate cannot be reproduced by the declared PRIMAT large-network implementation, if abundance feedback or post-target retuning is required to obtain the coordinate, if the live $A = 7$ response misses the preregistered survival tolerance, or if D/H, Y_p , or $^3\text{He}/\text{H}$ exceed their frozen disturbance controls. The observed lithium endpoint is downstream of that scientific decision and cannot rescue a failed gate.

The direct-reaction observable layer fails if the fixed zero-energy CPTG normalization or the declared external finite-energy/thermal carrier must be retuned reaction-by-reaction to recover the reported comparison rows. The first-order reaction-response layer fails if the frozen CPTG source direction cannot be pushed through the named native-network variational dynamics inside the preregistered vector, direction, conservation, and cross-solver gates. No success in these layers establishes a code-independent static endpoint matrix.

The late-lensing layer is evaluated only in its declared compressed coordinates. A galaxy-shear comparison cannot be promoted into a CMB-lensing claim, or vice versa, by mixing their definitions. For DESI, the compressed ShapeFit/BAO coordinate claim and the raw full-shape claim remain separate: raw full-shape validation requires nuisance-preserving AP, RSD, tracer-window, bias, counterterm, stochastic, and covariance machinery.

16.1 Sector Status Matrix

Each sector compares a fixed CPTG object to the observational coordinate, likelihood summary, or compressed variable appropriate to that sector. Status labels are intentionally matched to the strength of the implemented test.

Table 27: CPTG comparison status by sector.

Sector	CPTG comparison object	Status
Cosmological branch architecture	$\mathfrak{B}_\pi \rightarrow \{\mathfrak{B}_{\text{ac}}, \mathfrak{B}_{\text{lum}}\}$	One native branch; acoustic and luminosity branches are projections

Sector	CPTG comparison object	Status
Bound-system/native-phase interface	$a_\star, R_C, a_\pi, R_C, \eta_\star, H_{\text{struct}}, \mathcal{K}_\star$	Cross-scale acceleration/radius/rate identities defined; fixed-local-state, native-locking, and physical-evolution trajectories kept distinct
Background expansion	$E_\pi(a)^2, E_{\text{ac}}(a)^2, H_{\text{lum}}^{\text{anchor}}(a)$	Native background and exact acoustic projector defined; SH0ES-conditioned luminosity comparison history defined
Age and clock map	$t_\pi(a), t_{\text{ac}}(a), t_{\text{lum}}^{\text{anchor}}(a)$	Branch-specific clock integrals defined at common phase; no cross-branch common clock imposed
Luminosity comparison	$\mathcal{L}_{\text{loc}}, \Xi_{\text{lum}}^{\text{anchor}}(a), t_{0,\text{lum}}^{\text{anchor}}$	Phase-dependent SH0ES-conditioned projection defined; 10/9 retained only as an endpoint diagnostic candidate
Acoustic–luminosity midpoint	$\omega_{\text{AL}} = 1/2, \mathcal{M}_{\text{AL}}(a), H_{0,\text{mid}}^{\text{CPTG}}$	Symmetric midpoint prescription registered; rate-space result is data-conditioned; no third branch or independent fit parameter
Cosmological horizon	$\chi_\star, L_\star/R_C, z_{\text{enter}}$	Common-phase horizon coordinates registered; $L_\star/R_C = 0.53792156 < 1$ at last scattering
Supernovae	Standardized distance residuals with marginalized intercept	Full-covariance shape pass
BAO	$D_M/r_d, D_H/r_d, D_V/r_d, \text{AP coordinates}$	Coordinate comparison supported; fitted absolute normalization declared; no native absolute-ruler derivation claimed
Quarter-ruler	$\mathcal{G}_T^{-1/4}$	Supported wrapper diagnostic; independent derivation open
BBN background	$\Theta_{\text{BBN}}^{\text{CPTG}} = (\eta_{10}^{\text{BBN}}, N_{\text{eff}}^{\text{CPTG}}, \tau_7)$, with background read from the gate-off native row	PRIMAT v0.3.2 large-network full-coordinate validation pass; D/H and Y_p within preregistered controls
Nuclear reaction observable	$S_{12}^{\text{CPTG}}(0) = K_{S0} I_{C7}, S_{12}^{\text{CPTG}}(E) = S_{12}^{\text{CPTG}}(0) F_{\text{ext}}(E), \mathcal{R}_{pD}^{\text{CPTG}}(T)$	Commissioned $D(p, \gamma)^3\text{He}$ projection; 13/13 Mossa/LUNA S rows and 17/17 rate rows inside published intervals; external shape carrier declared
Nuclear reaction response	$\mathcal{C}_{r,\mathcal{N}}^{(1)} = \mathbf{P}_3 \mathbf{L}_{\mathcal{N}}^{(1)} \mathcal{A}_{r,\mathcal{N}}^{(1)} \mathbf{s}_r$	Network-conditioned first-order pushforward; PRIMAT 48/48 primary resolved tests pass, ParthENoPE supplies independent Reaction-20 transfer
Mass-seven abundance transport	$\tau_7 = \ln \pi, S_7 = 1/\pi, \text{live } ^7\text{Li}/^7\text{Be sink}$	23/23 scientific gates pass; measured survival 0.318366498376680; lithium endpoint downstream diagnostic
Galaxy weak lensing	$S_{8,\text{late}}^{\text{CPTG}} = \mathcal{G}_T S_{8,\text{anchor}}$	DES Y6/HSC/KiDS compressed support; maximum reported $ \text{pull} = 0.429317\sigma$
CMB lensing	$S_{8,\text{CMBL}}^{\text{CPTG}}$	ACT DR6 compressed-coordinate diagnostic support
CMB profiles	Locked acoustic-transport row	Profile-validation pass; exact gate closure
CMB posterior sampling	Same locked row	Likelihood-stability diagnostic; conventional convergence not claimed
CMB curvature-transport spectrum bridge	$C_T = \Phi_\pi/\Phi_0, \mathcal{R}_T = P_\pi/P_0 = C_T^2$	Route B Option 1 bridge locked; native perturbation/source implementation not claimed
CMB real-map envelope	$\mathcal{E}_\ell^M = (C_\ell^M/C_\ell^{\text{obs}})^{1/2}$, phase-locked $a_{\ell m}^M$	Planck/WMAP comparison-map closure pass; observed phase scaffold declared
DESI ShapeFit	Compressed ShapeFit vector	Coordinate-level pass
DESI full-shape spectrum	AP/growth spectrum-shell response	Completed exploratory diagnostic; raw nuisance-preserving full-shape validation not claimed

Each comparison layer therefore has a fixed object and a documented operational failure condition. A distance-sector implementation fails if the fixed distance relation misses covariance-level supernova or BAO structure after the appropriate coordinate map is applied. The primordial-network implementation fails if the prospectively frozen baryon or radiation coordinate cannot be reproduced natively, if the live mass-seven response misses the preregistered survival tolerance, or if the gate disturbs D/H, Y_p , or $^3\text{He}/\text{H}$ beyond the

frozen controls. The observed lithium endpoint remains downstream of that decision. Growth/lensing status is assessed against the appropriate compressed coordinate rather than by mixing galaxy-shear and CMB-lensing definitions.

For DESI, compressed-coordinate failure and raw full-shape failure remain separate: the former concerns the locked ShapeFit/BAO mappings; the latter requires the full nuisance-preserving AP, RSD, tracer-window, bias, counterterm, stochastic, and covariance machinery.

17 Limitations and Open Closure Conditions

The comparison program is intentionally modular, and several boundaries remain open by construction. First, the curvature-transport spectrum bridge fixes and validates the amplitude-to-power mapping but does not yet provide a completed native CPTG Boltzmann/source evolution law for $C_T(a, k)$, including full perturbation transport through the exact acoustic projector. The scalar metric, effective Poisson, and Weyl relations used in the comparison section remain external translation coordinates rather than CPTG-native GR/FLRW dynamics. Second, the BAO quarter-ruler relation remains a supported coordinate-wrapper candidate whose independent geometric derivation is not closed in this article. Third, the luminosity comparison history is SH0ES conditioned. Equations (127)–(130) define the phase-dependent luminosity/time comparison used here, while $\mathcal{G}_L = 10/9$ remains only an endpoint diagnostic candidate and is not required to define that history. The conditioned comparison realization is not promoted into a microscopic luminosity projector. Fourth, $E_{ac,\rho}$ is retained only as a density-weighted diagnostic overlay and is not another acoustic branch. Fifth, the DESI spectrum-shell result is a completed exploratory diagnostic with resolved tracer/multipole structure; a raw nuisance-preserving full-shape likelihood validation is outside the present claim scope and is not asserted here. Sixth, the compressed ShapeFit adapter holds $d_m = 0$; a nonzero spectral-shape coordinate requires an explicit CPTG derivation before it can be promoted. Seventh, the reported CMB posterior runs establish likelihood stability but do not meet the conventional Cobaya convergence criterion, so no conventional posterior-convergence claim is made. Eighth, the native π branch does not impose a universal phase-evolution law for the object-dependent structural acceleration $a_\star$: fixed local structural invariance and native locking are distinct comparison trajectories whose empirical discrimination requires structurally controlled bound systems across independent native phases, preferably with resolved multi-radius inference. Ninth, the acoustic–luminosity midpoint uses the fixed symmetric prescription $\omega_{AL} = 1/2$. Its realized midpoint values, including $\mathcal{M}_{AL}$, inherit the registry status and observational conditioning of their endpoints; neither the prescription nor its outputs define a third cosmological branch or an independently fitted Hubble parameter.

These open conditions do not alter the stronger claims already closed within their declared scopes: fixed-row CMB profile validation, Pantheon+ distance-shape comparison, compressed lensing support, the PRIMAT full-coordinate primordial-network test, and the frozen PRIMAT first-order source/Jacobian pushforward. They instead define the conditions that must be satisfied before those comparison layers can be promoted to stronger native or full-likelihood claims.

18 Likelihood, Software, and Evidence Provenance

The comparison workflow distinguishes external scientific sources, method/software sources, and CPTG-generated evidence outputs. An external citation establishes the data, convention, or method against which a calculation is performed; it does not by itself validate a CPTG-generated χ^2 , residual, or response vector. The major claim lanes are therefore bound explicitly below.

Table 28: Paper-wide provenance matrix. Artifact names identify the evidence lane; cryptographic identities for sealed bound-system and primary nuclear authority lanes are given in the Data, Code, and Evidence Availability section.

Claim lane	Upstream data / authority / artifact	Freeze or conditioning rule	Claim boundary
Native pi branch	CPTG framework and public computational materials [38, 47]	Native branch fixed before sector comparisons	Native/derived geometric quantities only
Bound-system/native-phase interface	NMBS-C7447 native-locking, local-invariance, and direct comparison analyses; XMM-VID1-2075 resolved-inference measurement test [43, 44, 45, 46]	Native phase is selected independently of the bound-system solve; each comparison trajectory and measurement interface is declared before cross-scale ratios are interpreted	Defines the cross-scale identities and measurement hierarchy; does not establish a universal phase-evolution law for a_*
Acoustic-luminosity midpoint	Registered weight $\omega_{AL} = 1/2$ applied to the acoustic and SH0ES-conditioned luminosity endpoints	Endpoint quantities are evaluated at the same N_π before midpoint formation; no independent midpoint fit	Symmetric comparison prescription; realized midpoint inherits endpoint conditioning and is not a third branch
CMB fixed-row likelihood tests	Planck, NPIPE/CamSpec, SPT-3G, ACT likelihood/data products and Cobaya machinery [2, 3, 37, 35, 36, 33, 34, 30, 32, 31]	CPTG row and gates fixed before each likelihood evaluation	Profile validation plus stated stability diagnostics; conventional MCMC convergence not claimed where $R - 1$ is too large
Route B spectrum bridge	CPTG Route B evidence chain [41, 47]	Squared-amplitude response mode fixed before bridge stress tests	Bridge validation only; no completed native Boltzmann/source evolution law
CMB real-map envelope	CMB comparison-map closure analysis [42, 47]	Observed harmonic phase scaffold declared and fixed	Data-conditioned map-space closure; no native random-phase prediction
Primordial baryon/radiation + $A = 7$ transport	PRIMAT v0.3.2 primary authority [14]; CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_AuthorityLock_20260819_2359EDT_r01.zip; companion lithium analysis [23]	Preregistration and radiation mapping frozen before target abundance exposure; 23 scientific gates fixed in advance	Tested PRIMAT large-network configuration only; lithium endpoint downstream
Direct $D(p, \gamma)^3\text{He}$ observable projection	CPTG commissioning paper, Solar Fusion II, Mossa/LUNA and comparison datasets [15, 16, 22, 17, 18, 19, 20, 21]	Zero-energy CPTG normalization fixed; finite-energy/thermal external carrier declared	Validates the commissioned observable projection; external energy shape is not native CPTG geometry
First-order nuclear reaction pushforward	PRIMAT evidence bundle CPTG_PRIMAT_PaperReadyValidationEvidence_20260817_r02.zip; accepted six-anchor source/Jacobian archive CPTG_PRIMAT_SIXANCHOR_NATIVE_SOURCE_JACOBIAN_RESULTS_20260817-202839.zip; PArthENoPE Reaction-20 authority bundle CPTG_v129_r109_PArthENoPE_CPTG_Reaction20_ReplicationAuthorityBundle_20260731_r02.zip [22, 24]	Source/Jacobian predictions frozen before authoritative endpoint coefficients are opened	Network-conditioned push-forward; no universal code-independent endpoint matrix
Pantheon+ distance shape	Pantheon+ data and covariance [5]	CPTG distance projection fixed; one intercept marginalized by declared protocol	Full-covariance distance-shape claim, not an H_0 calibration
Weak/CMB lensing compressed coordinates	DES Y6, HSC-Y3, KiDS-1000, ACT DR6, Planck anchor [26, 29, 27, 28, 33, 2]	Named anchor/coordinate fixed before residual calculation; overlapping analyses not combined as independent measurements	Compressed-coordinate support only
DESI ShapeFit / BAO / spectrum shell	DESI DR1 products and desilike conventions [7, 8, 9, 10, 11, 12]	Adapter choices and frozen ruler inputs declared before each comparison	ShapeFit coordinate pass; BAO coordinate comparison supported with fitted absolute normalization declared; quarter-ruler strongly supported as a diagnostic with independent derivation open; full-shape spectrum-shell retained as an exploratory diagnostic

Cobaya’s default MCMC stopping criterion is $R - 1 = 0.01$, and the documentation emphasizes that this diagnostic is necessary but not sufficient for convergence [31]. The posterior sections therefore report their

actual $R - 1$ values rather than treating job completion as conventional convergence.

CPTG-generated masks, likelihood outputs, scan tables, native-network result tables, and numerical summaries are bound to the supporting evidence releases and checksum identities described below where available [47]. Pipeline outputs whose release artifact has not yet been assigned a cryptographic identity remain labeled as pipeline results rather than being promoted into the parameter registry.

19 Reference Summary and Usage Boundaries

This guide defines the common technical language of the CPTG geometric pi branch. Its cosmological architecture is fixed: there is one native π branch, and the acoustic and luminosity branches are projections of it. The branches share the native phase $N_\pi = \ln a$, while each defined rate history generates its own integrated time through $H_b dt_b = dN_\pi$. No cross-branch thermal synchronization, second native background, GR/FLRW governing dynamics, or dark-matter sector is introduced. Its primary function is to keep downstream CPTG calculations and papers synchronized: the native branch is fixed first; redshift-selected native phase is kept distinct from any locally inferred bound-system $a_\star$; derived branch quantities and comparison coordinates follow from the stated dependency chains; the acoustic–luminosity midpoint uses the registered symmetric weight $\omega_{\text{AL}} = 1/2$, while each realized midpoint inherits the status of its endpoints; frozen external inputs and fixed implementation settings retain their separate provenance classes; transported, data-conditioned, abundance-transport, network-conditioned, and bridge-validated quantities are used only within their declared scopes.

For practical use, the parameter registry should be treated as the first lookup table and the sector construction sections as the corresponding how-to instructions. The nuclear and primordial sections specify the baryon, radiation, abundance-transport, and reaction-response coordinates. The CMB sections specify the exact acoustic background projection, acoustic transport row, amplitude and visibility gates, perturbation bridges, and CMB reference quantities. The late-time sections specify growth, lensing, distance, common-phase horizon, branch-specific clock, phase-dependent luminosity/time, BAO, and ShapeFit coordinates. The implementation and verification records then show how those definitions were exercised in the named numerical pipelines and native networks without changing their registry status.

This article does not use cross-sector agreement as a proof of CPTG and should not be cited as though it does. Sector-specific derivations, observational tests, likelihood analyses, and native-network studies are documented in their dedicated CPTG papers and evidence releases. The role of this reference is narrower and more durable: to state what each CPTG quantity is, where it comes from, how it is calculated or transported, what may be held fixed, what may not be reinterpreted, and which implementation record applies when the quantity is used elsewhere.

20 Data, Code, and Evidence Availability

External survey data, likelihood implementations, numerical software, and observational conventions are cited at their collaboration or software sources throughout this article. CPTG-generated calculations are associated with named evidence archives, result objects, and checksum identities where an immutable archival release has been established. Repository locations serve as distribution mirrors; for cryptographically sealed evidence lanes, the archive filename together with its SHA-256 digest defines the immutable evidence identity [47].

20.1 Bound-System Native-Phase and Resolved-Inference Evidence

The bound-system/native-phase interface is supported by the two NMBS-C7447 comparison trajectories and by the XMM-VID1-2075 resolved-inference measurement test. These archives document distinct questions and are not interchangeable evidence authorities.

NMBS-C7447 native-locking evidence

CPTG_NMBS-C7447_NativeLock_Structural_Evidence_20260904_r04_FINAL_EVIDENCE.zip

SHA-256: fc3076d3ee5a83fd17f1a1a75fe73844bb36974440a0a204d7009c446247420a

NMBS-C7447 local-structural-invariance evidence

CPTG_NMBS-C7447_LocalStructuralInvariance_Evidence_20260904_r01_FINAL_EVIDENCE.zip

SHA-256: 875c079e44e6ce4e5963987591bd7dfc8e4b7d36ecee046ab45b743702b44a2

XMM-VID1-2075 full-Jeans structural-inference evidence

XMM-VID1-2075_FullJeans_aStar_Inversion_Test_20260904.zip

SHA-256: d6571f12fd815ef94fa70891a494c4ae9b87dfcb07145c1dbb7750d414887455

The NMBS archives preserve the two deliberately different native-phase continuation tests from the same observed state. The XMM archive preserves the resolved Jeans inversion used to test how the inferred a_* and its downstream cross-scale ratios depend on the measurement interface. Together they support the reference-level distinction between phase selection, phase continuation, and structural inference [45, 46].

20.2 PRIMAT Full-Coordinate Primordial-Network Evidence

The PRIMAT baryon, radiation, and mass-seven full-coordinate validation is bound by the following authority lock.

Authority lock

CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_AuthorityLock_20260819_2359EDT_r01.zip

SHA-256

43b65036e229dc5f04d445f5890399236aaf546c598f1711e572043db6b05bec

Recognized final-evidence archive

CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_20260819_2344EDT_r05_FINAL_EVIDENCE.zip

SHA-256

9c44543cd2497e72170c0a64173423c9b3bf2f1f3924c9d4e77c16a318fb8b4c

The authority lock binds the full-coordinate evidence lane to the recognized final-evidence archive and preserves the preregistered baryon/radiation mapping, native-network execution, mass-seven transport result, scientific gates, and associated audit record.

20.3 PRIMAT First-Order Reaction-Response Evidence

The PRIMAT first-order network-conditioned reaction-response validation is preserved separately from the full-coordinate mass-seven evidence.

Primary validation archive

CPTG_PRIMAT_PaperReadyValidationEvidence_20260817_r02.zip

SHA-256

b41c8ee49477d8330deb56c85a230d8dcc1d4bba80a1405251a4d6ea7f9b3205

Frozen prediction object SHA-256

679e3e6fc7432c0592004b9bf0985b0756323ceb327050f3303ca6c816f31bff

Accepted six-anchor source/Jacobian results

CPTG_PRIMAT_SIXANCHOR_NATIVE_SOURCE_JACOBIAN_RESULTS_20260817-202839.zip

SHA-256

8511f3f79d4298d077a32625859440e01f3b530a3eda6a3e999582e1f938baf8

These identities bind the frozen source/Jacobian predictions to the accepted six-anchor native endpoint-response evaluation and preserve the separation between prediction construction and authoritative endpoint-reference exposure.

20.4 PArthENoPE Second-Authority Transfer Evidence

The independent PArthENoPE Reaction-20 transfer used as the second numerical authority within its declared shared-endpoint scope is preserved in the following authority bundle.

PArthENoPE authority bundle

CPTG_v129_r109_PArthENoPE_CPTG_Reaction20_ReplicationAuthorityBundle_20260731_r02.zip

SHA-256

3d42d1cb1d710248841db8d7b1ceafcc7569f092b9b35c9f4114ce9615074cfc

This archive preserves the independent network-transfer evidence cited for the PArthENoPE comparison and does not broaden that result beyond the Reaction-20 transfer scope defined in the text.

20.5 CMB, Supernova, Lensing, and DESI Evidence

The CMB, Pantheon+, weak-lensing, CMB-lensing, ShapeFit, BAO, and DESI spectrum-shell results are supported by the cited observational data products, likelihood implementations, CPTG-generated masks, scan tables, likelihood outputs, residual calculations, and numerical summaries described in their respective sections.

These quantities remain classified as *pipeline results* because they are outputs of the declared comparison and likelihood procedures rather than native CPTG invariants or frozen theory parameters. Where a separate cryptographically sealed release is available, its checksum identity may additionally be used to verify the exact distributed artifact. The presence or absence of a separate archival checksum does not change the evidentiary status of the reported calculation or its stated claim boundary.

20.6 Evidence Identity and Distribution

For cryptographically sealed evidence lanes, reproducibility is anchored by three elements:

1. the named evidence archive or result object;
2. its SHA-256 digest; and
3. the corresponding calculation, freeze rule, or validation scope stated in this article.

The public CPTG repository and associated distribution locations provide access to the supporting materials [47]. Repository paths are distribution mechanisms rather than scientific identities. Where a checksum-bound archive is listed above, the archive filename and SHA-256 digest provide the immutable reference for independently obtained copies.

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